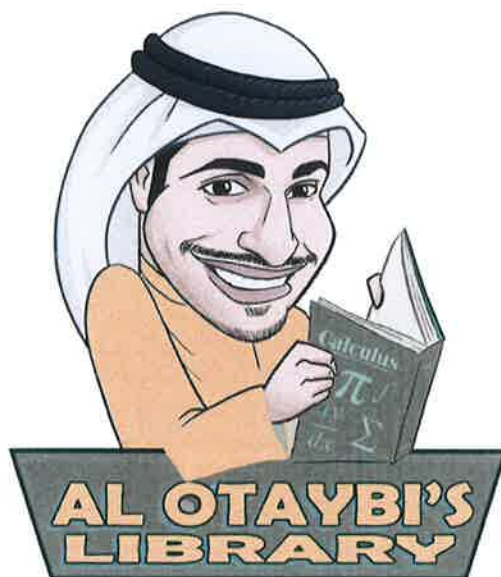


Kuwait University

2021/2022

Spring course



Calculus B

Final

Sections : 10.4 (part b/b)

10.4 Areas and Lengths in Polar Coordinates.

(18)

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi



10.4 Area in polar coordinate



$$A = \frac{1}{2} \int_a^b r^2 d\theta$$

دالتين

دالة وحدة

inside both

inside one and
outside the
another

تطبيق مباشر
للقانون

$$A_T = A_1 + A_2$$

$$A_T = A_1 - A_2$$



خطوات الحل



إذا عندي دالتين
ايجاد حدود التقاطع

$$r = r$$

- (1) الرسم
- (2) تحديد المساحة المطلوبة
- (3) بحث الـ Symmetry
- (4) ايجاد حدود التكامل

شكل مألوف
دوائر / قلوب

شكل غير مألوف

(1) أعطي الـ Symmetry

إذا ما ضبطت البداية
أحول لحد التقاطع



(3) أوقف عند حد التقاطع و
نهاية المساحة

(1) أحط ورقتي عند
بداية ونهاية
الـ Area

(2) أساوي النقاط
بالـ r

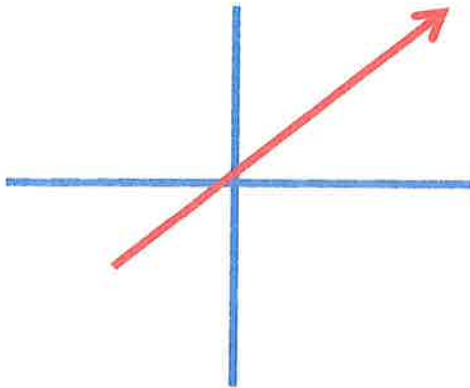


Graphs

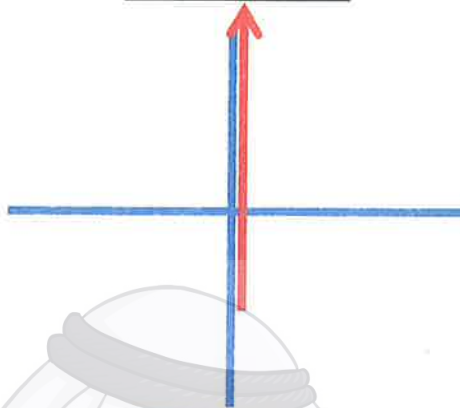


Line

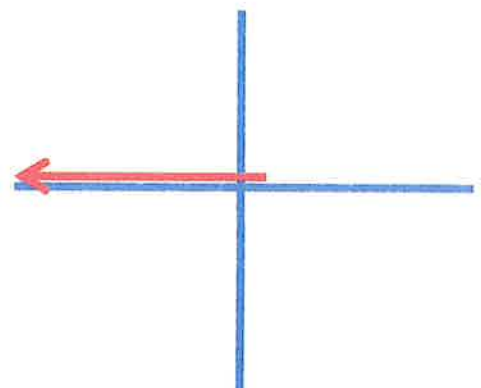
$$\theta = \frac{\pi}{4}$$



$$\theta = \frac{\pi}{2}$$

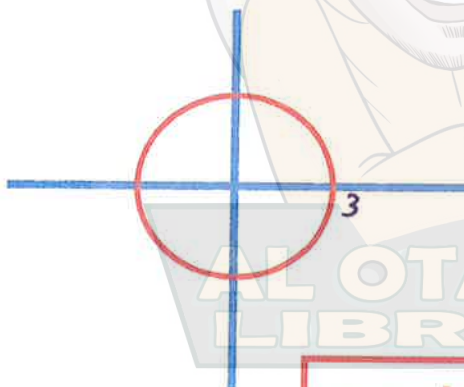


$$\theta = \pi$$

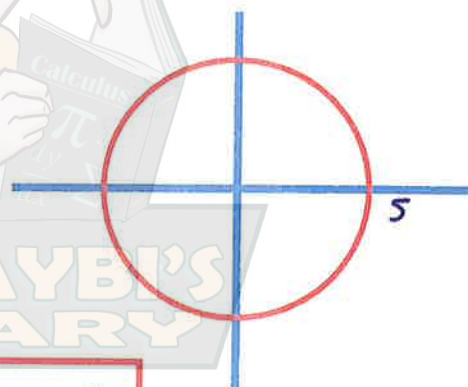


Circle

$$r = 3$$

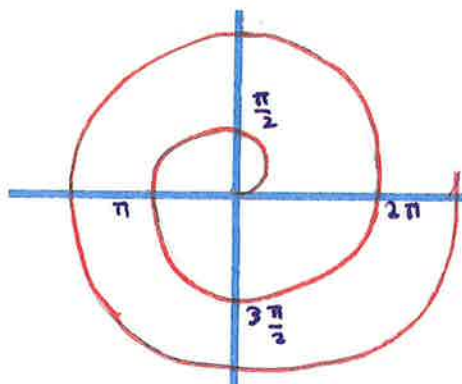


$$r = 5$$



Spiral of arch

$$r = \theta$$

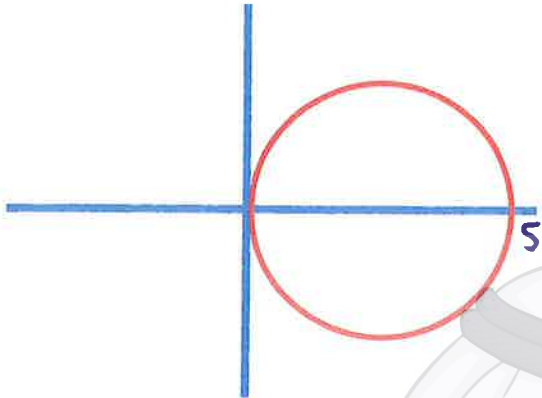




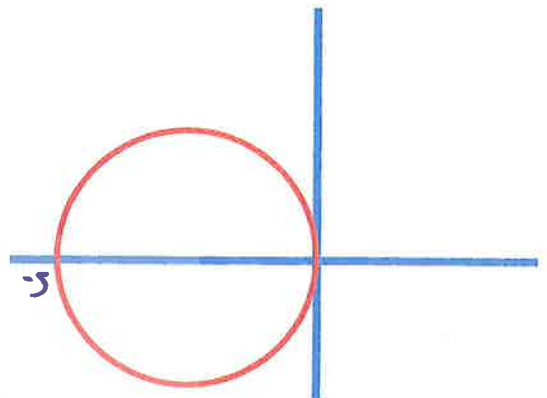
Circles



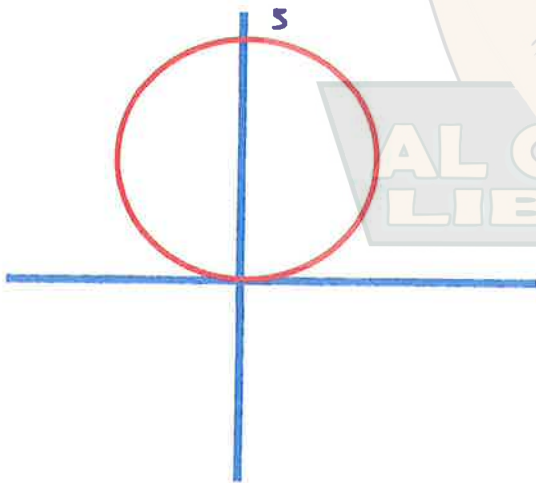
$$r = 5\cos\theta$$



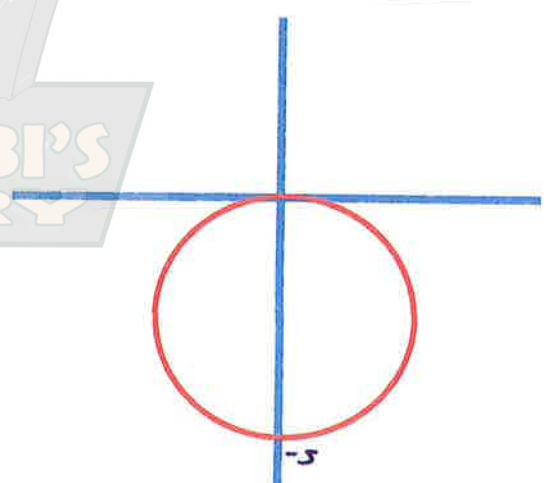
$$r = -5\cos\theta$$



$$r = 5\sin\theta$$



$$r = -5\sin\theta$$



AL OTAYBI'S
LIBRARY

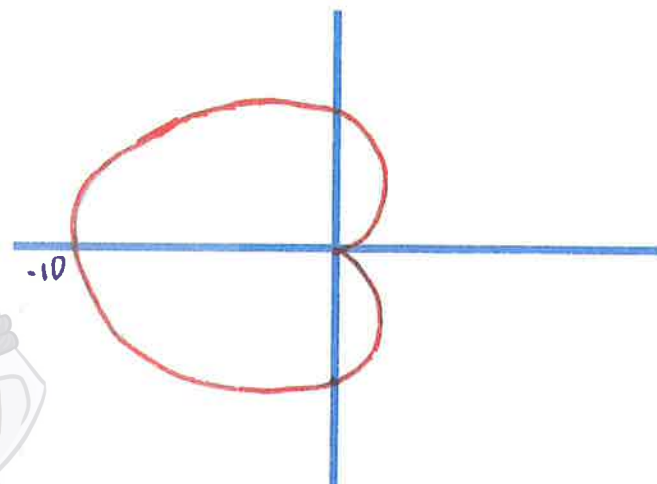
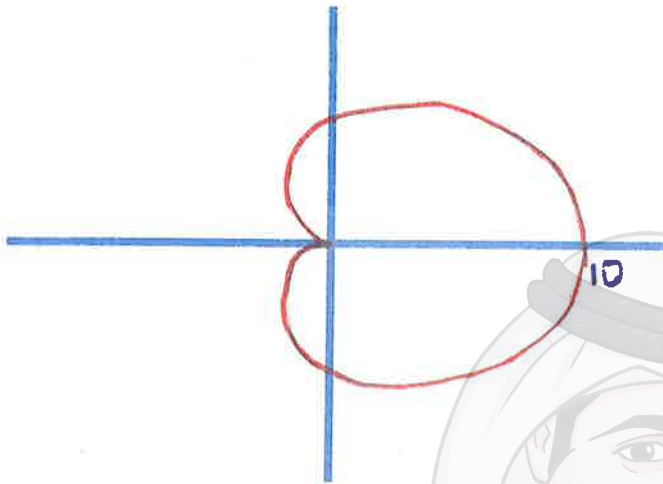


Cardioid



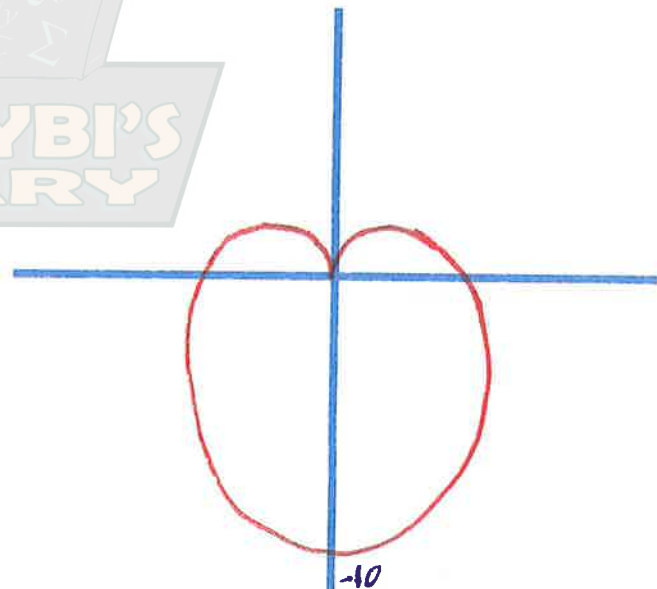
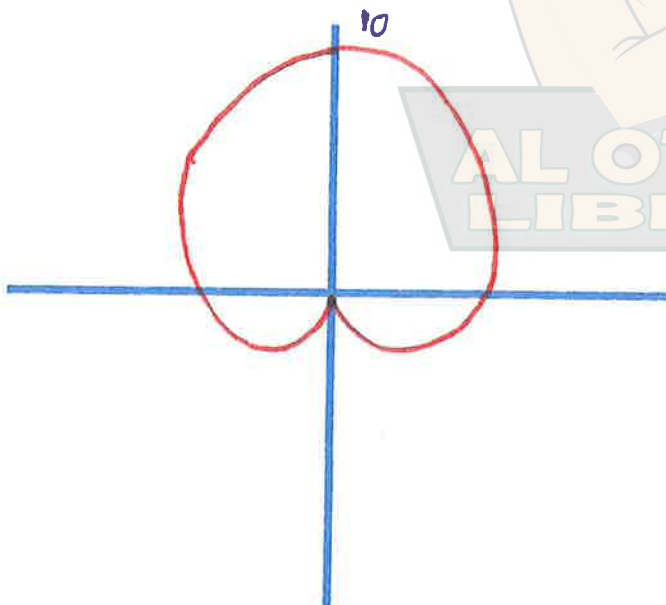
$$r = 5 + 5\cos\theta$$

$$r = 5 - 5\cos\theta$$



$$r = 5 + 5\sin\theta$$

$$r = 5 - 5\sin\theta$$

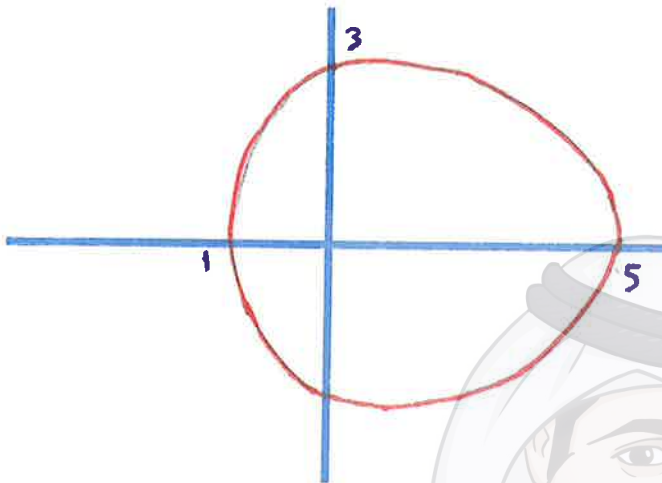




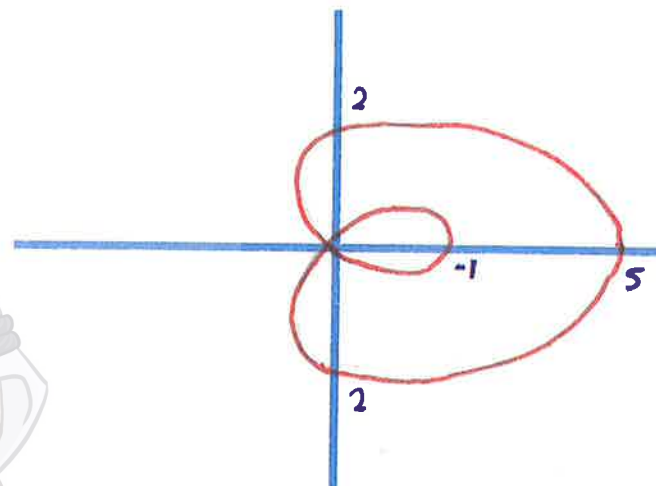
Limacons



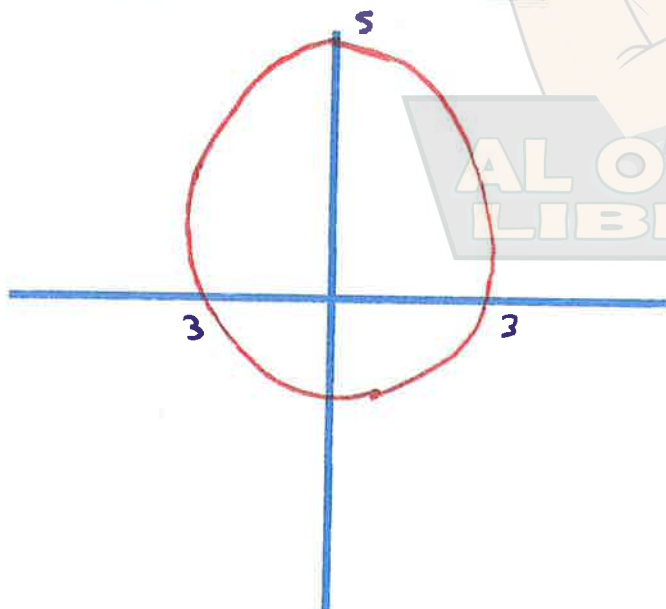
$$r = 3 + 2\cos\theta$$



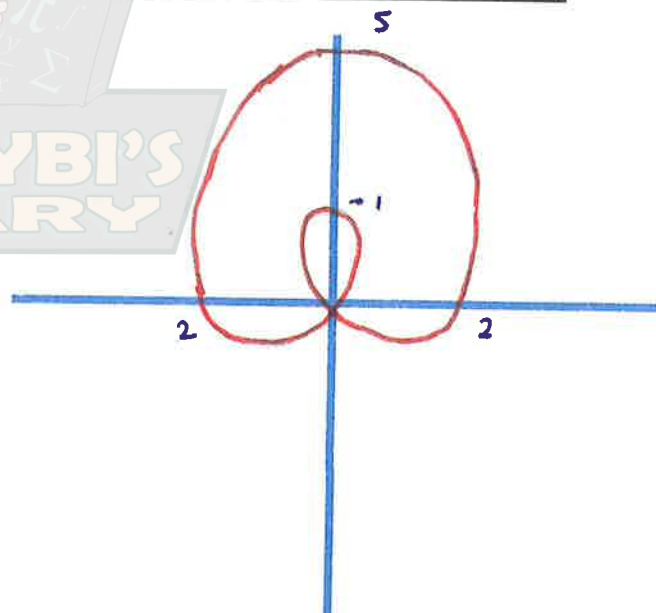
$$r = 2 + 3\cos\theta$$



$$r = 3 + 2\sin\theta$$



$$r = 2 + 3\sin\theta$$

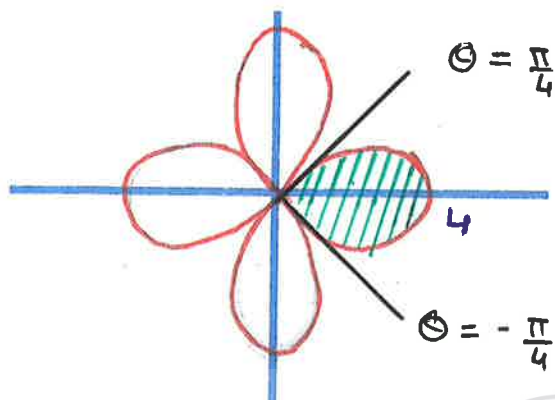




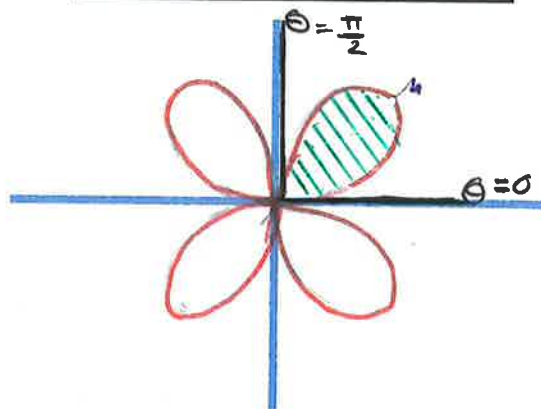
Leaved Roses



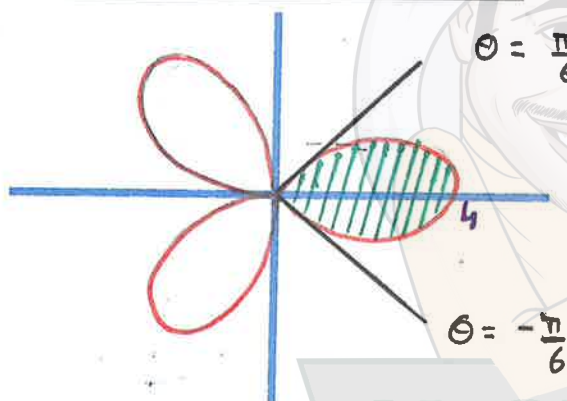
$$r = 4\cos(2\theta)$$



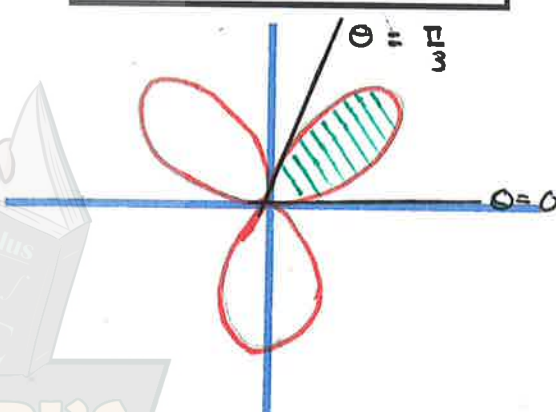
$$r = 4\sin(2\theta)$$



$$r = 4\cos(3\theta)$$

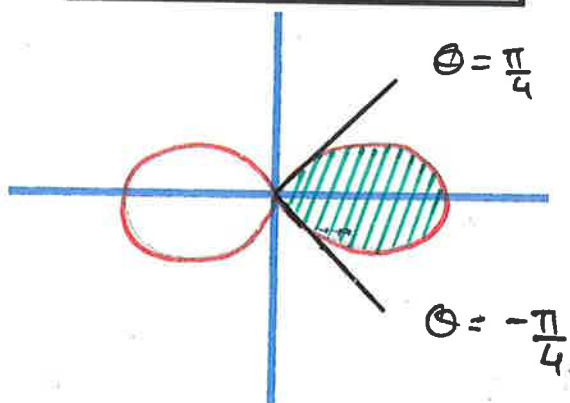


$$r = 4\sin(3\theta)$$



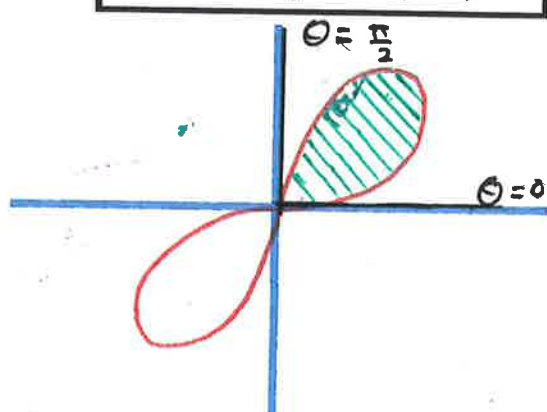
$$r^2 = 4\cos(2\theta)$$

$$r = 2\sqrt{\cos(2\theta)}$$



$$r^2 = 4\sin(2\theta)$$

$$r = 2\sqrt{\sin(2\theta)}$$



AL OTAYBI'S
LIBRARY

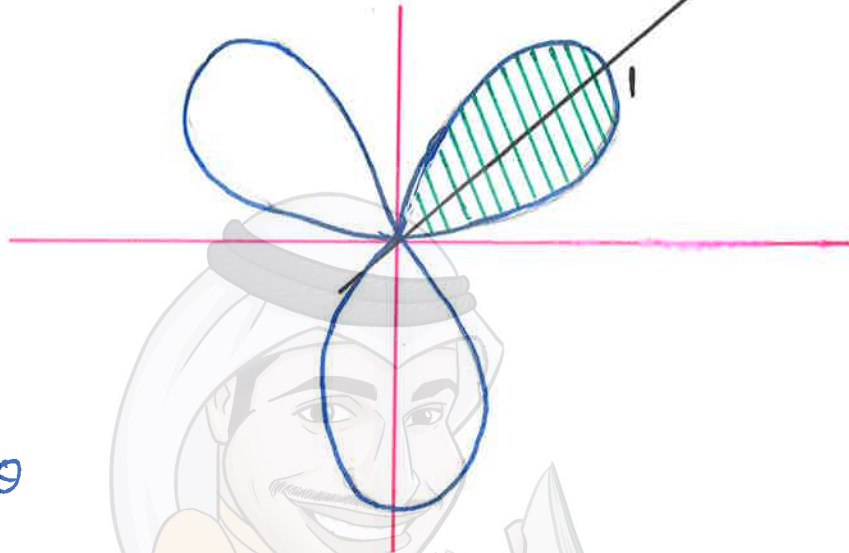
Find the area enclosed by one loop of the curve

$$r = \sin(3\theta)$$



Solution 2

الحل مع Symmetry



$$A = \frac{1}{2} \int r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} \sin^2(3\theta) d\theta$$

Symmetry

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} [1 - \cos(6\theta)] d\theta$$

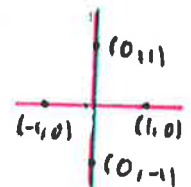
$$= \frac{1}{2} \left[\theta - \frac{\sin 6\theta}{6} \right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{6} - \frac{\sin \pi}{6} \right) - 0 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right] = \frac{\pi}{12}$$

$$0 = \sin 3\theta$$

$$1 = \sin 3\theta$$



$$3\theta = 0, 3\theta = \pi$$

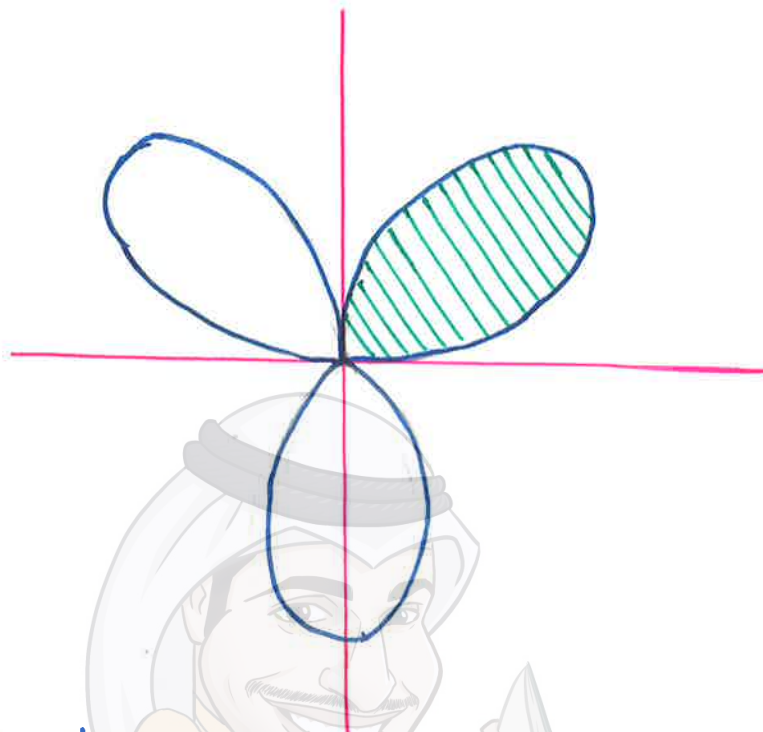
$$\theta = 0, \theta = \frac{\pi}{3}$$

$$3\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{6}$$

Find the area enclosed by one loop of the curve

$$r = \sin(3\theta)$$



$$A = \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin^2(3\theta) d\theta$$

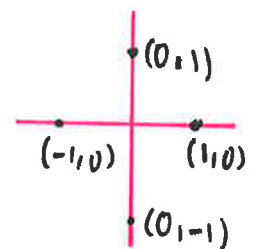
$$= \frac{1}{2} \cdot \frac{1}{2} \int_0^{\frac{\pi}{3}} [1 - \cos(6\theta)] d\theta$$

$$= \frac{1}{4} \left[\theta - \frac{\sin(6\theta)}{6} \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{4} \left[\left(\frac{\pi}{3} - \frac{\sin 2\pi}{6} \right) - 0 \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{3} - 0 \right] = \frac{\pi}{12}$$

$$0 = \sin 3\theta$$



$$3\theta = 0$$

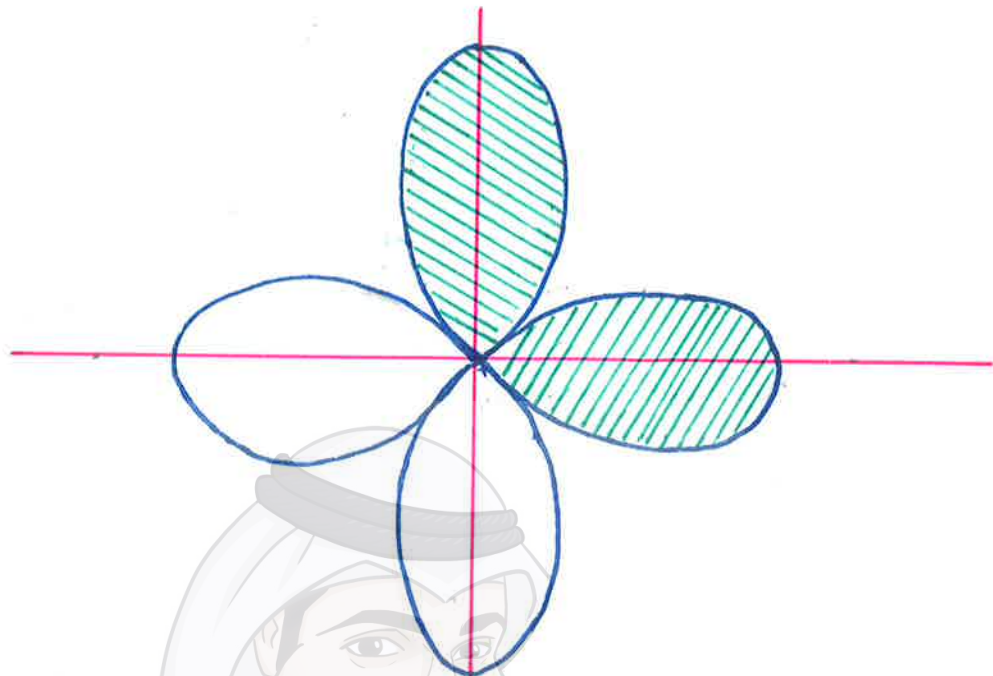
$$\theta = 0$$

$$3\theta = \pi$$

$$\theta = \frac{\pi}{3}$$

Find the area enclosed by two loops of the polar curve

$$r = \cos 2\theta$$



Symmetry

$$A = 2 \int_{-\pi/4}^{\pi/4} \frac{1}{2} r^2 d\theta$$

$$= \int_{-\pi/4}^{\pi/4} \cos^2 2\theta d\theta$$

$$= \frac{1}{2} \int_{-\pi/4}^{\pi/4} (1 + \cos 4\theta) d\theta$$

$$= \frac{1}{2} \left[\theta + \frac{\sin 4\theta}{4} \right]_{-\pi/4}^{\pi/4}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{4} + \frac{\sin \pi}{4} \right) - \left(-\frac{\pi}{4} + \frac{\sin(-\pi)}{4} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} + 0 + \frac{\pi}{4} \right]$$

$$= \frac{1}{2} \cdot \frac{2\pi}{4} = \frac{\pi}{4}$$

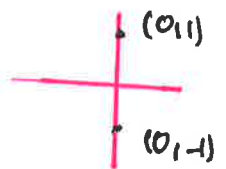
$$0 = \cos 2\theta$$

$$2\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{4}$$

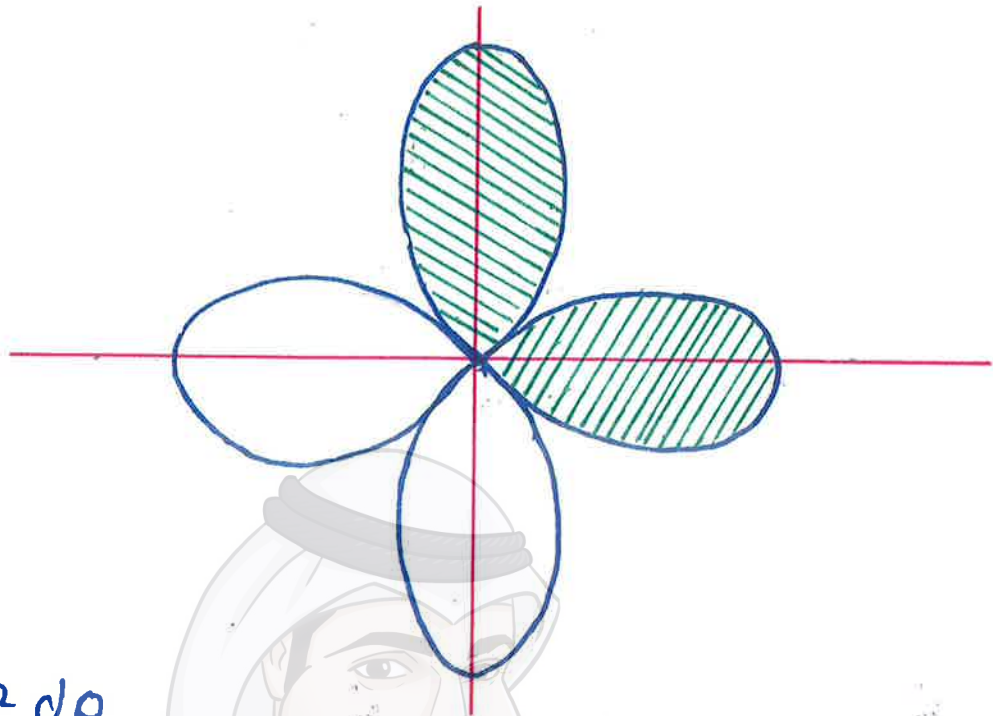
$$2\theta = -\frac{\pi}{2}$$

$$\theta = -\frac{\pi}{4}$$



Find the area enclosed by two loops of the polar curve

$$r = \cos 2\theta$$



Symmetry

$$A = 4 \int_0^{\frac{\pi}{4}} r^2 d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} \cos^2 2\theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{4}} (1 + \cos 4\theta) d\theta$$

$$= \left[\theta + \frac{\sin 4\theta}{4} \right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{\pi}{4} + \frac{\sin \pi}{4} \right) - (0 + 0)$$

$$= \left[\frac{\pi}{4} + 0 \right]$$

$$= \frac{\pi}{4}$$

Given the polar curve

$$r = \frac{\pi}{2} - 2\theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

a) Sketch the curve.

b) Find the area of the region the region enclosed by the curve and above the x-axis

(a) $r = \frac{\pi}{2} - 2\theta$

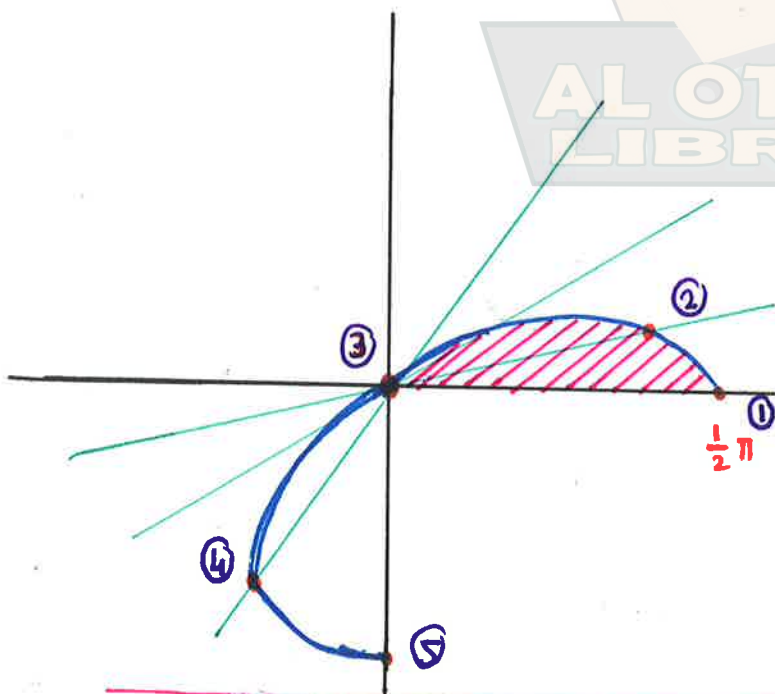
θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
r	$\frac{1}{2}\pi$	$\frac{1}{6}\pi$	0	$-\frac{1}{6}\pi$	$-\frac{1}{2}\pi$

$$\begin{aligned} \frac{\pi}{2} - 2 \cdot \frac{\pi}{6} &= \frac{\pi}{2} - \frac{\pi}{3} \\ &= \frac{3\pi}{6} - \frac{2\pi}{6} \\ &= \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} \frac{\pi}{2} - 2 \cdot \frac{\pi}{4} &= \frac{\pi}{2} - \frac{\pi}{2} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \frac{\pi}{2} - 2 \cdot \frac{\pi}{3} &= \frac{3\pi}{6} - \frac{4\pi}{6} \\ &= -\frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} \frac{\pi}{2} - 2 \cdot \frac{\pi}{2} &= \frac{\pi}{2} - \pi \\ &= -\frac{\pi}{2} \end{aligned}$$



$$\begin{aligned} 0 &= \frac{\pi}{2} - 2\theta \\ 2\theta &= \frac{\pi}{2} \\ \theta &= \frac{\pi}{4} \end{aligned}$$

* لتقدير الحدود *

$$\begin{aligned} \frac{\pi}{2} &= \frac{\pi}{2} - 2\theta \\ 0 &= -2\theta \\ \theta &= 0 \end{aligned}$$

$$A = \frac{1}{2} \int r^2 d\theta$$

$$= \frac{1}{2} \int \left(\frac{\pi}{2} - 2\theta \right)^2 d\theta$$

$$= \frac{1}{2} \frac{\left(\frac{\pi}{2} - 2\theta \right)^3}{(-2) \cdot 3}$$

$$= \frac{1}{-12} \left[\left(\frac{\pi}{2} - 2\theta \right)^3 \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{-12} \left[0 - \left(\frac{\pi}{2} \right)^3 \right] = \frac{1}{12} \frac{\pi^3}{8}$$

$$= \frac{\pi^3}{96}$$

Let C be the polar curve

$$r = \sqrt{\theta} \cos \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

a) Sketch the curve.

b) Find the area enclosed by C

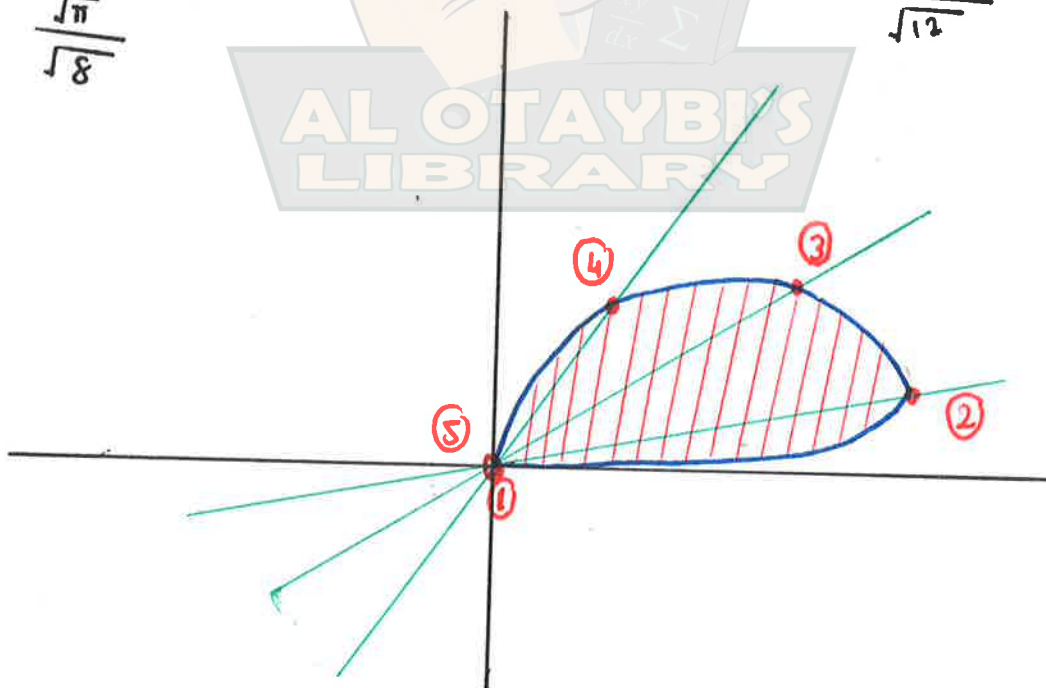
$$r = \sqrt{\theta} \cos \theta$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
r	0	$\frac{1}{\sqrt{8}}\sqrt{\pi}$	$\frac{1}{\sqrt{8}}\sqrt{\pi}$	$\frac{1}{\sqrt{12}}\sqrt{\pi}$	0

$$\begin{aligned} & \sqrt{\frac{\pi}{6}} \cos \frac{\pi}{6} \\ & \frac{\sqrt{\pi}}{\sqrt{6}} \cdot \frac{\sqrt{3}}{2} \\ & \frac{\sqrt{\pi}}{\sqrt{2}} \cdot \frac{1}{2} \\ & \frac{\sqrt{\pi}}{\sqrt{2}} \cdot \frac{1}{\sqrt{4}} \\ & \frac{\sqrt{\pi}}{\sqrt{8}} \end{aligned}$$

$$\begin{aligned} & \sqrt{\frac{\pi}{4}} \cos \frac{\pi}{4} \\ & \frac{\sqrt{\pi}}{\sqrt{4}} \cdot \frac{1}{\sqrt{2}} \\ & \frac{\sqrt{\pi}}{\sqrt{8}} \end{aligned}$$

$$\begin{aligned} & \sqrt{\frac{\pi}{3}} \cos \frac{\pi}{3} \\ & \frac{\sqrt{\pi}}{\sqrt{3}} \cdot \frac{1}{2} \\ & \frac{\sqrt{\pi}}{\sqrt{3}} \cdot \frac{1}{\sqrt{4}} \\ & \frac{\sqrt{\pi}}{\sqrt{12}} \end{aligned}$$



$$A = \frac{1}{2} \int_0^{\frac{\pi}{2}} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\sqrt{\theta} \cos \theta)^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \theta \cos^2 \theta d\theta$$

أولوية بعد تطبيق قانون الدوّار

$$= \frac{1}{2} \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} \theta (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{4} \int_0^{\frac{\pi}{2}} (\theta + \theta \cos 2\theta) d\theta$$

$$\frac{\theta^2}{2} + \int \theta \cos 2\theta d\theta$$

$$u = \theta \quad \begin{aligned} dv &= \cos 2\theta d\theta \\ du &= d\theta \quad \leftarrow v = \frac{\sin 2\theta}{2} \end{aligned}$$

$$\frac{1}{2} \theta \sin 2\theta - \frac{1}{2} \int \sin 2\theta d\theta$$

$$\frac{1}{2} \theta \sin 2\theta + \frac{1}{2} \frac{\cos 2\theta}{2}$$

$$\frac{1}{4} \left[\frac{\theta^2}{2} + \frac{1}{2} \theta \sin 2\theta + \frac{1}{4} \cos 2\theta \right]_0^{\frac{\pi}{2}}$$

$$\frac{1}{4} \left[\left(\frac{(\frac{\pi}{2})^2}{2} + 0 + \frac{1}{4} (-1) \right) - \left(0 + 0 + \frac{1}{4} \right) \right]$$

$$\frac{1}{4} \left[\frac{\pi^2}{8} - \frac{1}{2} \right]$$

Find the area of the region that is inside the graphs of both equations

$$r = 2 + 2\cos\theta \quad \text{and} \quad r = 3$$

$$r = 2 + 2\cos\theta$$

$$r = 3$$

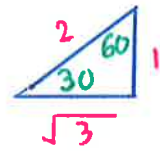
$$r = r$$

$$2 + 2\cos\theta = 3$$

$$2\cos\theta = 1$$

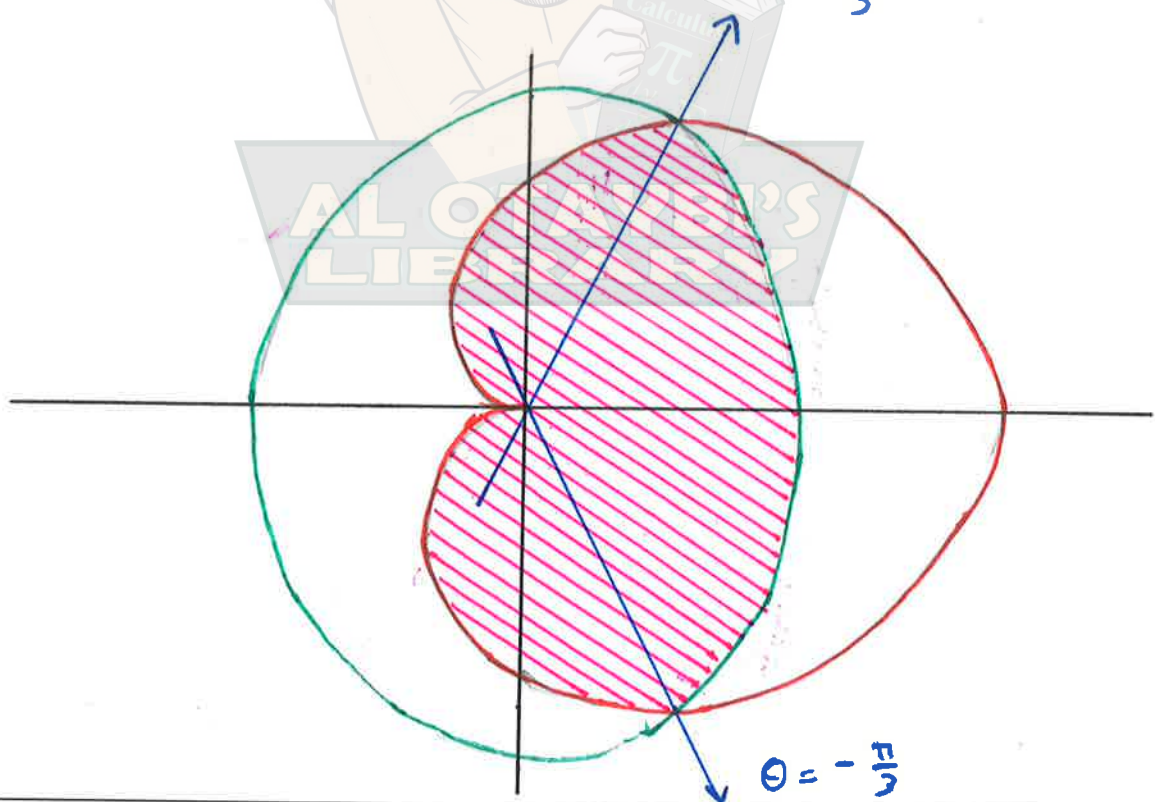
$$\div 2 \quad \cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}, \quad \theta = -\frac{\pi}{3}$$



$$\frac{\pi}{3}$$

S	A	α
t	C	$-\alpha$



$$A_1 = \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (2 + 2\cos\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (4 + 8\cos\theta + 4\cos^2\theta) d\theta$$

$$\downarrow$$

$$\frac{1}{2}(1 + \cos 2\theta)$$

$$= \frac{1}{2} \left[4\theta + 8\sin\theta + 2\left(\theta + \frac{\sin 2\theta}{2}\right) \right]_{\frac{\pi}{3}}^{\pi}$$

$$= \frac{1}{2} \left[\left(4\pi + 0 + 2(\pi + 0) \right) - \left(\left(\frac{4\pi}{3} + \frac{8\sqrt{3}}{2} \right) + 2\left(\frac{\pi}{3} + \frac{\frac{\sqrt{3}}{2}}{2} \right) \right) \right]$$

$$= \frac{1}{2} \left[6\pi - \left(\frac{6\pi}{3} + \frac{9\sqrt{3}}{2} \right) \right]$$

$$= \frac{1}{2} \left[4\pi - \frac{9\sqrt{3}}{2} \right] = 2\pi - \frac{9\sqrt{3}}{4}$$

$$\Rightarrow A_T = 2(A_1 + A_2)$$

$$= 2 \left(2\pi - \frac{9\sqrt{3}}{4} + \frac{3\pi}{2} \right)$$

$$= \pi - \frac{9\sqrt{3}}{2} + 3\pi$$

$$= 4\pi - \frac{9\sqrt{3}}{2}$$

$$A_2 = \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} (3)^2 d\theta$$

$$= \frac{1}{2} 9 \left[\theta \right]_0^{\frac{\pi}{3}}$$

$$= \frac{9}{2} \left[\frac{\pi}{3} - 0 \right]$$

$$= \frac{3\pi}{2}$$

a) Sketch the graph of the polar equation

$$r = 2 - 2\cos\theta \quad \text{and} \quad r = 2$$

b) Set up (but not evaluate) the integrals representing the area inside the curves

$$r = 2 - 2\cos\theta \quad \text{and} \quad r = 2$$

$$r = 2 - 2\cos\theta$$

$$r = 2$$

$$r = r$$

$$2 - 2\cos\theta = 2$$

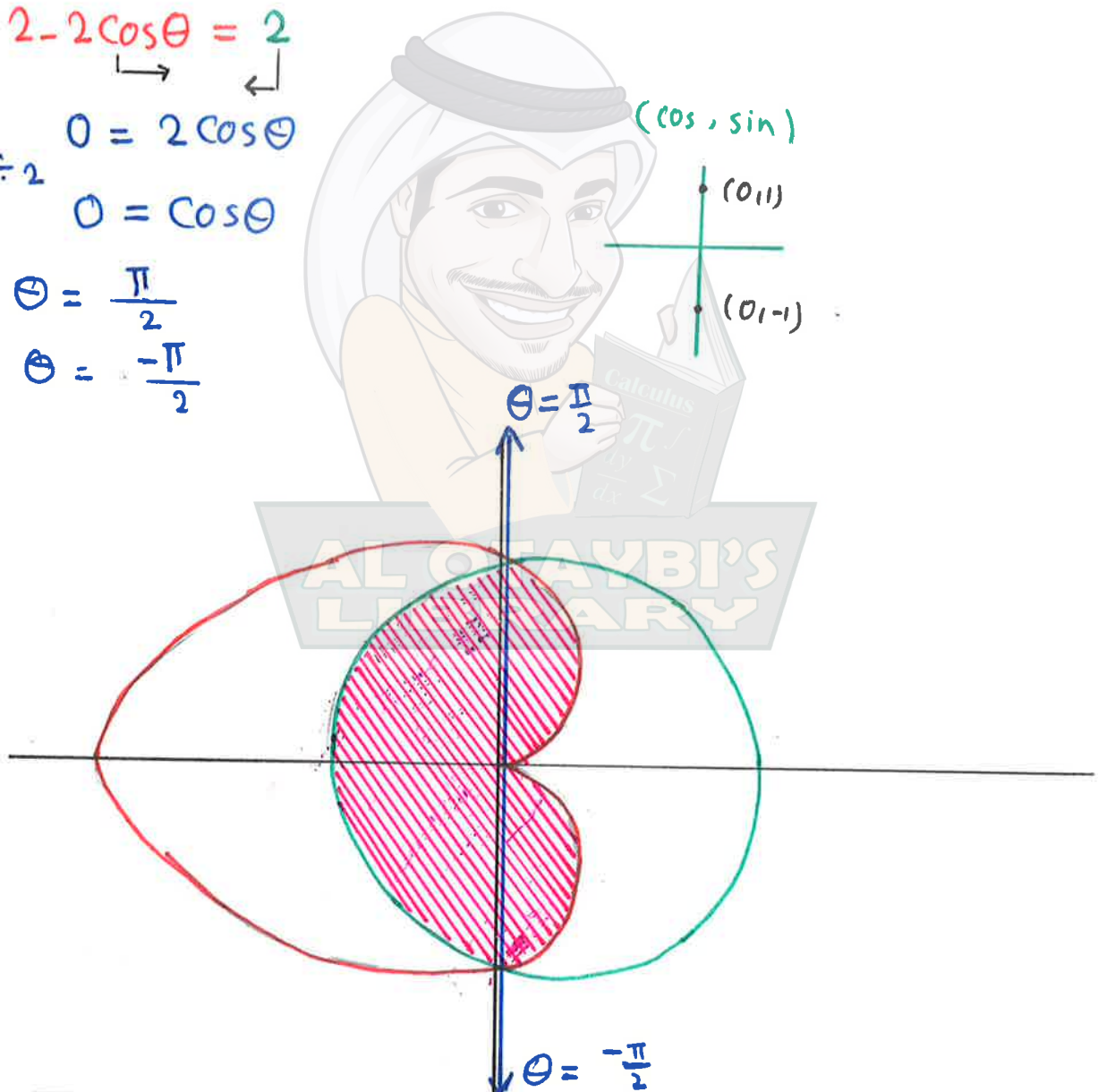
$$0 = 2\cos\theta$$

$\div 2$

$$0 = \cos\theta$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \theta = -\frac{\pi}{2}$$



$$A_1 = \frac{1}{2} \int_0^{\frac{\pi}{2}} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} (2 - 2\cos\theta)^2 d\theta$$

$$A_2 = \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} (2)^2 d\theta$$

$$\Rightarrow A_T = 2(A_1 + A_2)$$



Sketch the graph of the polar equation

$$r = 3 - 3\sin\theta \quad \text{and} \quad r = 1 + \sin\theta$$

and find the area of the region that lies inside both graphs.

$$r = 3 - 3\sin\theta$$

$$r = 1 + \sin\theta$$

$$r = r$$

$$3 - 3\sin\theta = 1 + \sin\theta$$

$$3 - 1 = \sin\theta + 3\sin\theta$$

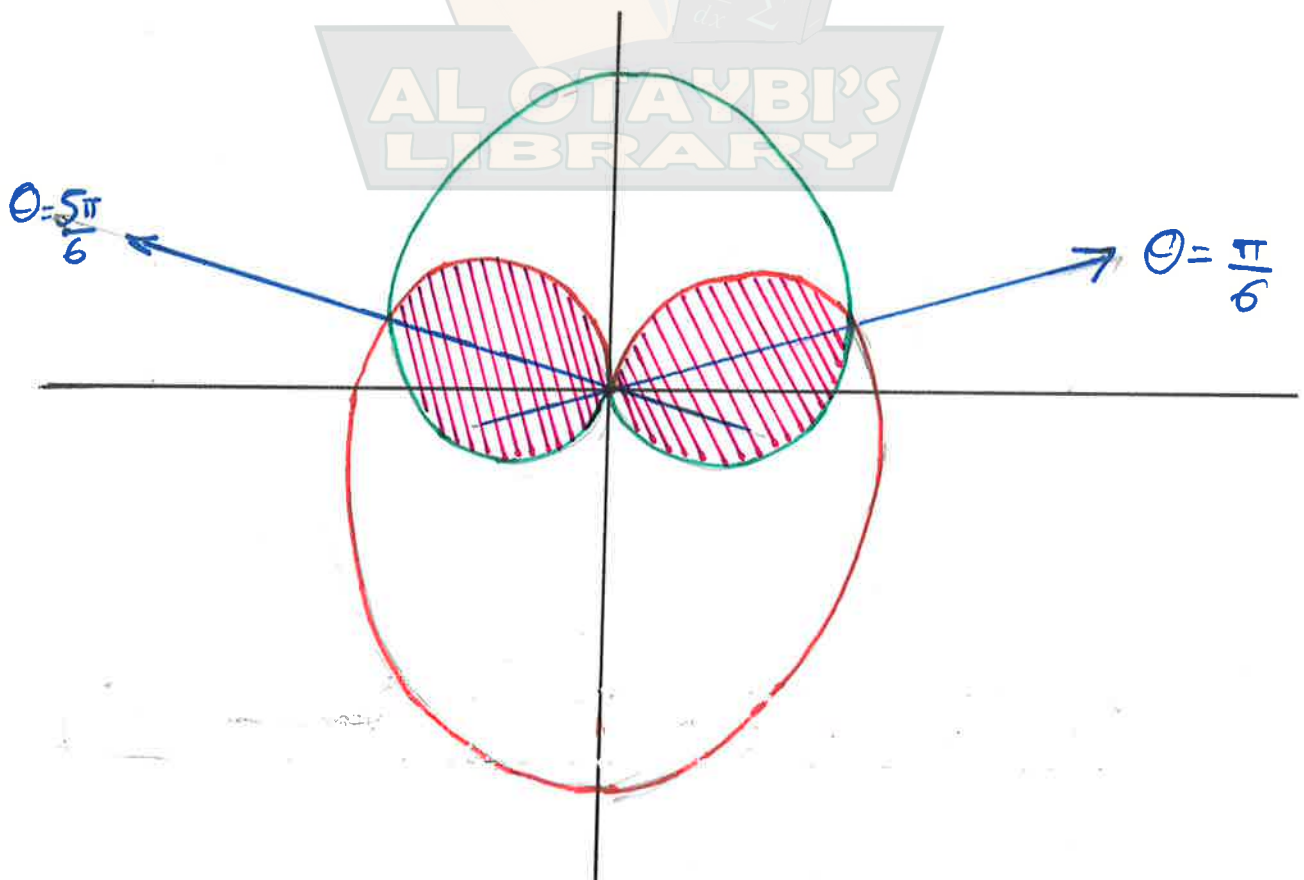
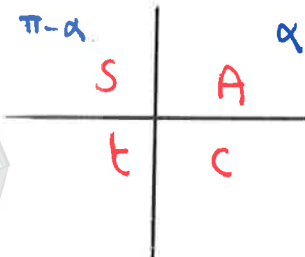
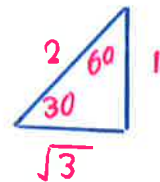
$$2 = 4\sin\theta$$

$$\sin\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\theta = \pi - \frac{\pi}{6} \Rightarrow \theta = \frac{5\pi}{6}$$

$$\frac{\pi}{6}$$



$$A_1 = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (3 - 3\sin\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (9 - 18\sin\theta + 9\sin^2\theta) d\theta$$

$$\frac{1}{2}(1 - \cos 2\theta)$$

$$= \frac{1}{2} \left[9\theta - 18\sin\theta + \frac{9}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\frac{9}{2} - 18 + \frac{9}{2} \left(\frac{\pi}{2} - 0 \right) \right) + \left(\frac{9\pi}{6} - 18\frac{1}{2} + \frac{9}{2} \left(\frac{\pi}{6} - \frac{\sin \pi}{2} \right) \right) \right]$$

$$A_2 = \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} r^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} (1 + \sin\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} (1 + 2\sin\theta + \sin^2\theta) d\theta$$

$$\frac{1}{2}(1 - \cos 2\theta)$$

$$= \frac{1}{2} \left[\theta - 2\cos\theta + \frac{1}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) \right]_{-\frac{\pi}{2}}^{\frac{\pi}{6}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{6} - 0 + \frac{1}{2} \left(\frac{\pi}{6} - 0 \right) \right) - \left(-\frac{\pi}{2} - 0 + \frac{1}{2} \left(-\frac{\pi}{2} - 0 \right) \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{2} + \frac{\pi}{4} - \left(-\frac{\pi}{2} - \frac{\pi}{4} \right) \right]$$

$$\frac{\pi}{2} + \frac{\pi}{4} + \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{2}$$

$$= \frac{1}{2} \cdot \frac{3\pi}{2} = \frac{3\pi}{4}$$

$$\Rightarrow A_T = 2(A_1 + A_2)$$

Find the area of the region that is inside the graphs of both polar equations

$$r = \sqrt{3}\cos\theta \quad \text{and} \quad r = \sin\theta$$

$$r = \sqrt{3}\cos\theta \quad r = \sin\theta$$

$$r = r$$

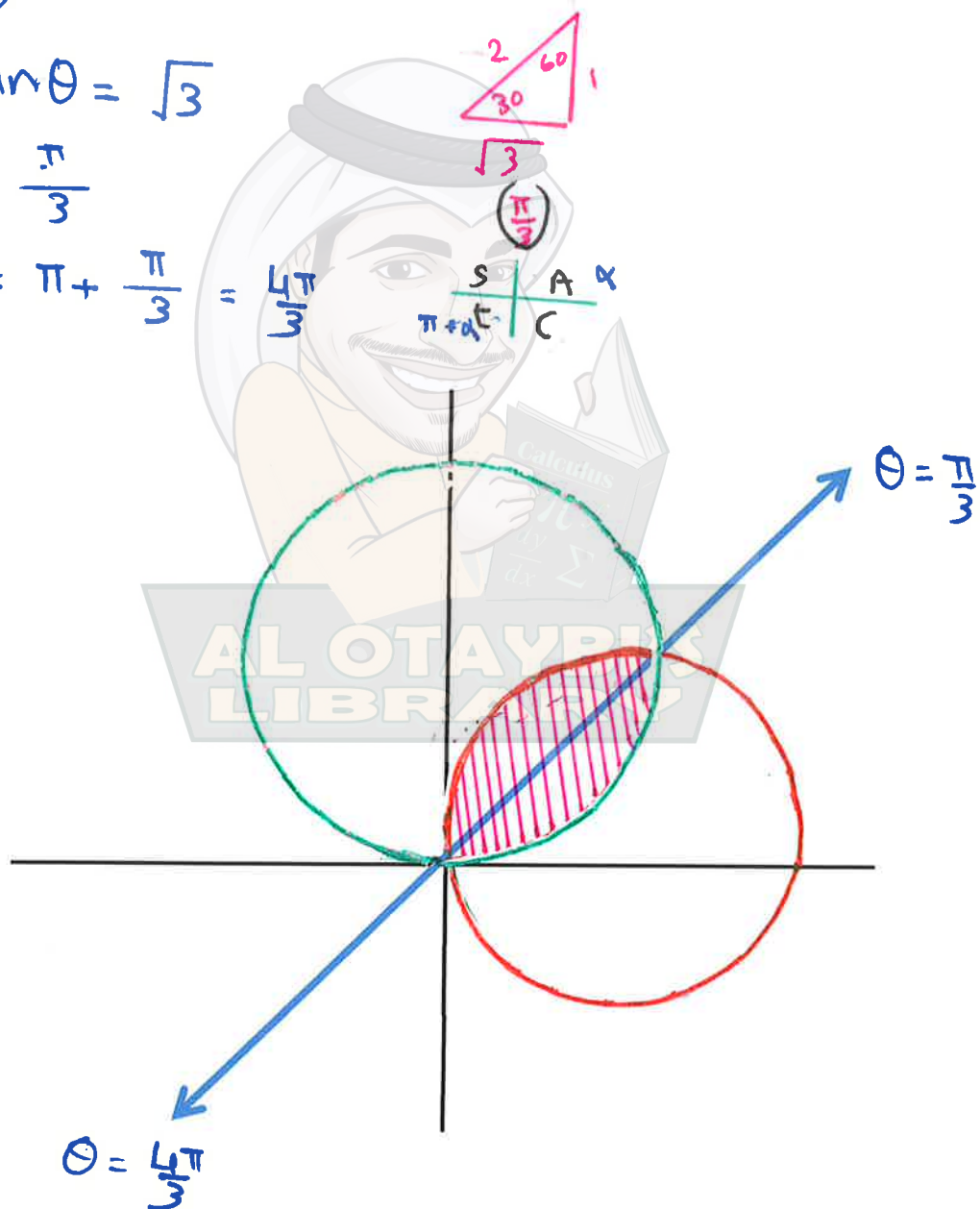
$$\sqrt{3}\cos\theta = \sin\theta$$

$$\div \cos\theta$$

$$\tan\theta = \sqrt{3}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

$$\Rightarrow \theta = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$



$$\begin{aligned}
 A_1 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} r^2 d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\sqrt{3} \cos \theta)^2 d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} 3 \cos^2 \theta d\theta \\
 &= \frac{3}{2} \cdot \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta \\
 &= \frac{3}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\
 &= \frac{3}{4} \left[\left(\frac{\pi}{2} + 0 \right) - \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \right] \\
 &= \frac{3}{4} \left[\frac{3\pi}{6} - \frac{2\pi}{6} + \frac{\sqrt{3}}{4} \right] \\
 &= \frac{3}{4} \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right]
 \end{aligned}$$

$$\begin{aligned}
 A_2 &= \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta \\
 &= \frac{1}{2} \cdot \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - \cos 2\theta) d\theta \\
 &= \frac{1}{4} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{3}} \\
 &= \frac{1}{4} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{2} - 0 \right] \\
 &= \frac{1}{4} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow A_T &= A_1 + A_2 \\
 &= \frac{3}{4} \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right] + \frac{1}{4} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] \\
 &= \frac{1}{4} \left[\frac{3\pi}{6} + \frac{3\sqrt{3}}{4} + \frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] \\
 &= \frac{1}{4} \left[\frac{5\pi}{6} - \frac{2\sqrt{3}}{4} \right]
 \end{aligned}$$



Nice Question !!



Find the area of the region that is inside
the graphs of equation

$$r = \sin\theta \quad \text{and} \quad r = \sin(2\theta)$$

$$r = \sin\theta$$

$$r = \sin(2\theta)$$

$$r = r$$

$$\sin\theta = \sin(2\theta)$$

$$\sin\theta = 2\sin\theta \cos\theta$$

$$2\sin\theta \cos\theta - \sin\theta = 0$$

$$\sin\theta [2\cos\theta - 1] = 0$$

$$\sin\theta = 0$$

$$2\cos\theta - 1 = 0$$

$$2\cos\theta = 1$$

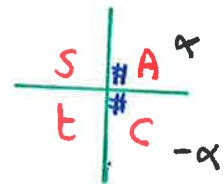
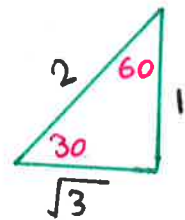
$$\Rightarrow \theta = 0$$

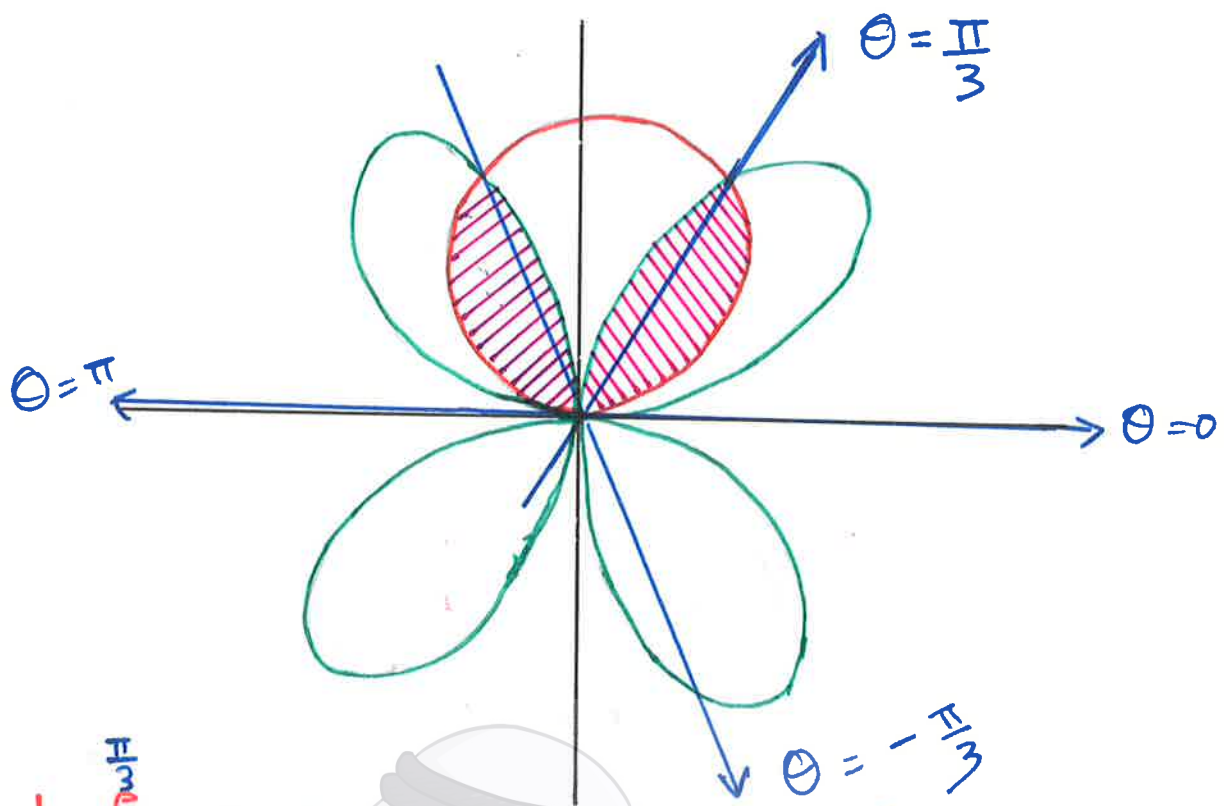
$$\cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

$$\Rightarrow \theta = -\frac{\pi}{3}$$

$$\Rightarrow \theta = \pi$$





$$\begin{aligned}
 A_1 &= \frac{1}{2} \int_0^{\pi/3} r^2 d\theta \\
 &= \frac{1}{2} \int_0^{\pi/3} \sin^2 \theta d\theta \\
 &= \frac{1}{2} \cdot \frac{1}{2} \int_0^{\pi/3} (1 - \cos 2\theta) d\theta \\
 &= \frac{1}{4} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/3} \\
 &= \frac{1}{4} \left[\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right) - 0 \right]
 \end{aligned}$$

$$\begin{aligned}
 A_2 &= \frac{1}{2} \int_{\pi/3}^{\pi} r^2 d\theta \\
 &= \frac{1}{2} \int_{\pi/3}^{\pi} \sin^2(2\theta) d\theta \\
 &= \frac{1}{2} \cdot \frac{1}{2} \int_{\pi/3}^{\pi} (1 - \cos 4\theta) d\theta \\
 &= \frac{1}{4} \left[\theta - \frac{\sin 4\theta}{4} \right]_{\pi/3}^{\pi} \\
 &= \frac{1}{4} \left[\left(\frac{\pi}{2} - 0 \right) - \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \right]
 \end{aligned}$$

$$A_T = 2(A_1 + A_2)$$

Find the area of the region that lies inside the polar curve $r = 1 - \sin\theta$ and outside the polar curve $r = 1$

$$r = 1 - \sin\theta \quad r = 1$$

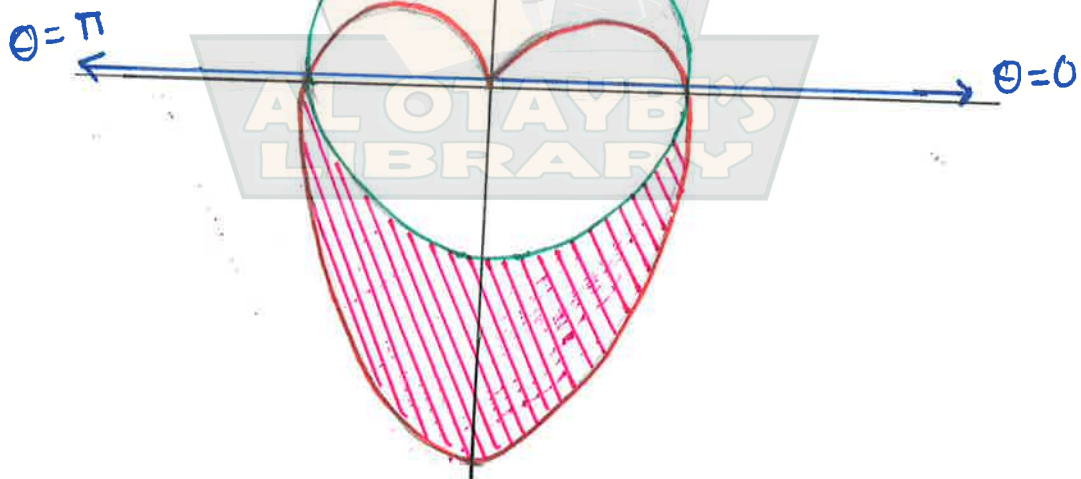
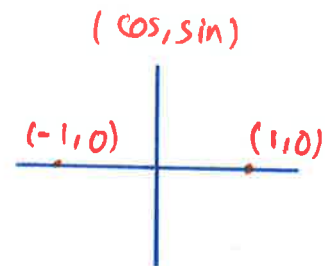
$$r = r$$

$$1 - \sin\theta = 1$$

$$0 = \sin\theta$$

$$\Rightarrow \theta = 0$$

$$\Rightarrow \theta = \pi$$



$$A_1 = \frac{1}{2} \int_{-\frac{\pi}{2}}^0 r^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^0 (1 - \sin\theta)^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^0 (1 - 2\sin\theta + \sin^2\theta) d\theta$$

$$\Downarrow$$

$$\frac{1}{2} (1 - \cos 2\theta)$$

$$= \frac{1}{2} \left[\theta + 2\cos\theta + \frac{1}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) \right]_{-\frac{\pi}{2}}^0$$

$$= \frac{1}{2} \left[(0 + 2 + \frac{1}{2}(0)) - \left(-\frac{\pi}{2} + 0 + \frac{1}{2} \left(-\frac{\pi}{2} - 0 \right) \right) \right]$$

$$= \frac{1}{2} \left[2 - \left(-\frac{2\pi}{4} - \frac{\pi}{4} \right) \right]$$

$$= \frac{1}{2} \left[2 + \frac{3\pi}{4} \right]$$



$$A_T = 2 (A_1 - A_2)$$

$$= 2 \left(\frac{1}{2} \left[2 + \frac{3\pi}{4} \right] - \frac{\pi}{4} \right)$$

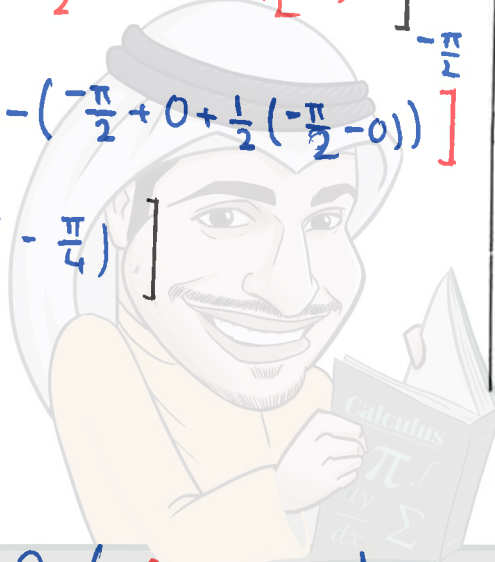
$$A_2 = \frac{1}{2} \int_{-\frac{\pi}{2}}^0 r^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^0 (1)^2 d\theta$$

$$= \frac{1}{2} [\theta]_{-\frac{\pi}{2}}^0$$

$$= \frac{1}{2} [0 - (-\frac{\pi}{2})]$$

$$= \frac{\pi}{4}$$



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Find the area of the region that lies inside the circle

$r = \frac{1}{2}$ and outside the cardioid $r = 1 - \cos\theta$

$r = \frac{1}{2}$

$r = 1 - \cos\theta$

$r = r$

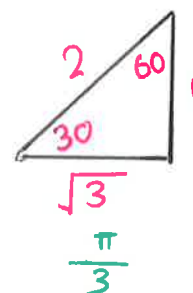
$\frac{1}{2} = 1 - \cos\theta$

$\cos\theta = 1 - \frac{1}{2}$

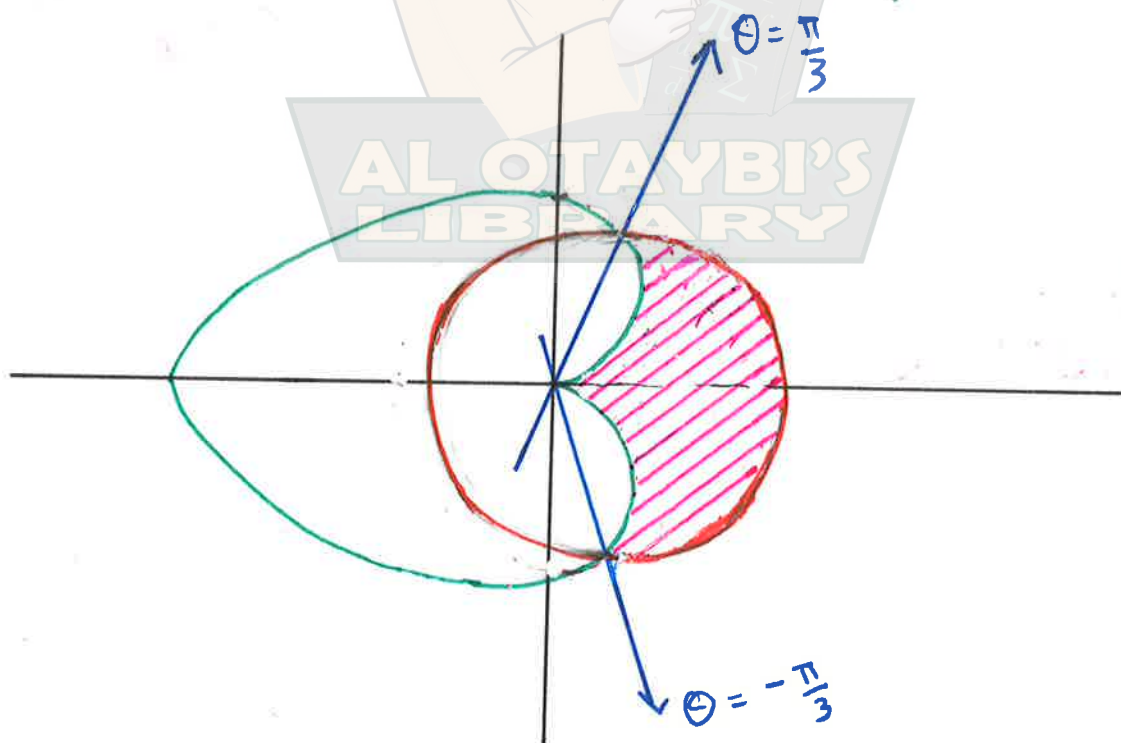
$\cos\theta = \frac{1}{2}$

$\Rightarrow \theta = \frac{\pi}{3}$

$\Rightarrow \theta = -\frac{\pi}{3}$



	#	α
S	A	
T	C	$-\alpha$
	#	



$$A_1 = \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} \left(\frac{1}{2}\right)^2 d\theta$$

$$= \frac{1}{2} \cdot \frac{1}{4} \left[\theta \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{8} \left[\frac{\pi}{3} - 0 \right]$$

$$= \frac{\pi}{24}$$

$$A_2 = \frac{1}{2} \int_0^{\frac{\pi}{3}} r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - \cos \theta)^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - 2\cos \theta + \cos^2 \theta) d\theta$$

$$\downarrow$$

$$\frac{1}{2} (1 + \cos 2\theta)$$

$$= \frac{1}{2} \left[\theta - 2\sin \theta + \frac{1}{2} \left(\theta + \frac{\sin 2\theta}{2} \right) \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left[\left(\pi - 0 + \frac{1}{2} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \right) - 0 \right]$$

$$\Rightarrow A_T = 2 (A_1 - A_2)$$

$$= 2 \left(\frac{\pi}{24} - \frac{1}{2} \left[\pi + \frac{1}{2} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \right] \right)$$

Find the area of the region that is inside the curve
 $r = 4(1 - \cos\theta)$ and outside the circle $r = 2$

$$r = 4 - 4\cos\theta$$

$$r = 2$$

$$r = r$$

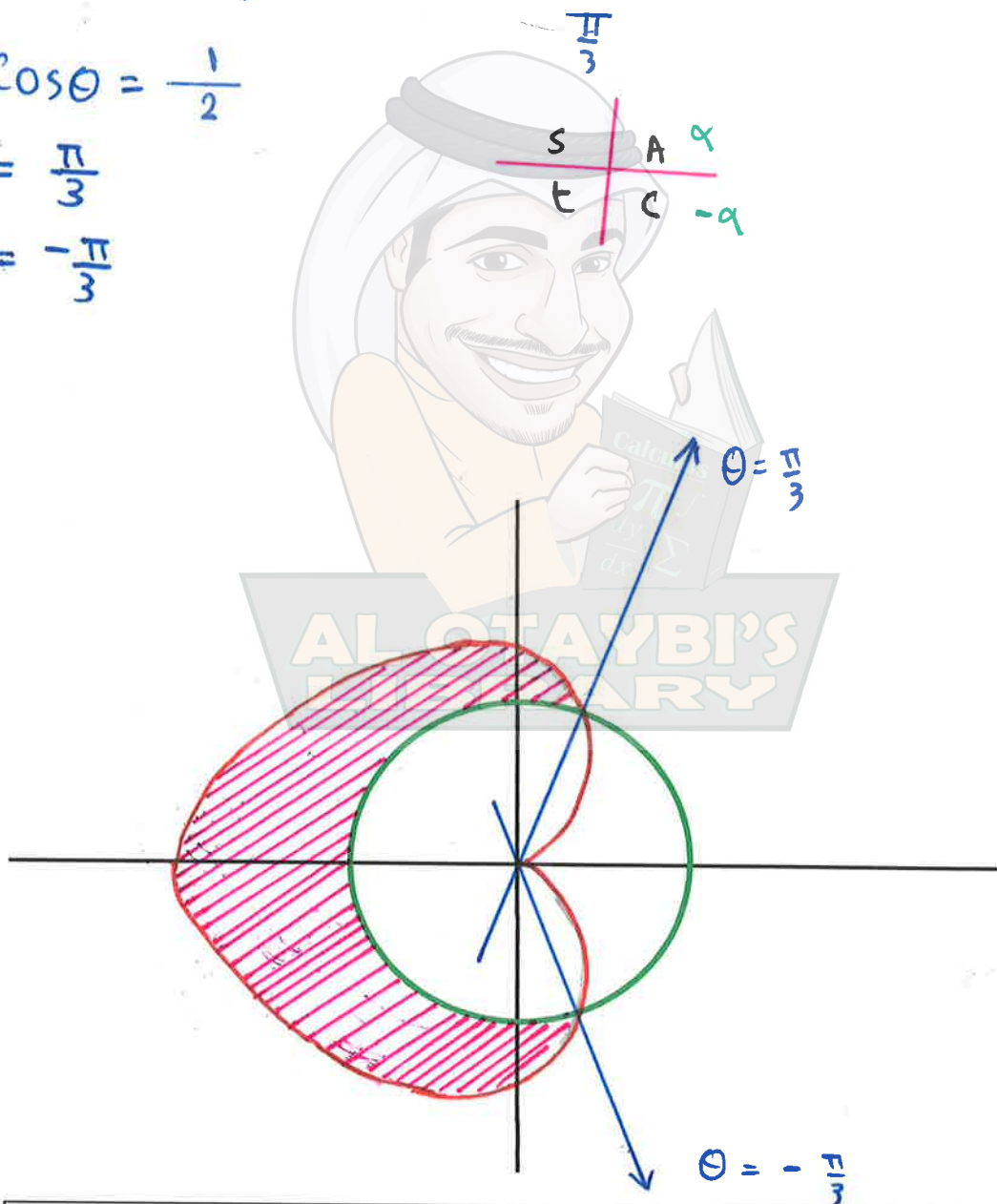
$$4 - 4\cos\theta = 2$$

$$2 = 4\cos\theta$$

$$\cos\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

$$\Rightarrow \theta = -\frac{\pi}{3}$$



$$\begin{aligned}
 A_1 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} r^2 d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (4 - 4\cos\theta)^2 d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (16 - 32\cos\theta + 16\cos^2\theta) d\theta \\
 &= \frac{1}{2} 16 \cdot \int_{\frac{\pi}{3}}^{\pi} (1 - 2\cos\theta + \cos^2\theta) d\theta
 \end{aligned}$$

$$\begin{aligned}
 &\quad \quad \quad \frac{1}{2} (1 + \cos 2\theta) \\
 &= 8 \left[\theta - 2\sin\theta + \frac{1}{2} \left(\theta + \frac{\sin 2\theta}{2} \right) \right]_{\frac{\pi}{3}}^{\pi} \\
 &= 8 \left[\left(\pi - 0 + \frac{1}{2} (\pi + 0) \right) - \left(\frac{\pi}{3} - \frac{2\sqrt{3}}{2} + \frac{1}{2} \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \right) \right] \\
 &= 8 \left[\frac{3\pi}{2} - \frac{3\pi}{6} + \frac{7\sqrt{3}}{8} \right] \\
 &= 8 \left[\pi + \frac{7\sqrt{3}}{8} \right]
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow A_T &= 2(A_1 - A_2) \\
 &= 2 \left(8 \left(\pi + \frac{7\sqrt{3}}{8} \right) - 2 \left(\pi - \frac{\pi}{3} \right) \right)
 \end{aligned}$$

$$\begin{aligned}
 A_2 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} r^2 d\theta \\
 &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\pi} (2)^2 d\theta \\
 &= \frac{1}{2} \cdot 4 \cdot [\theta]_{\frac{\pi}{3}}^{\pi} \\
 &= 2 \left[\pi - \frac{\pi}{3} \right]
 \end{aligned}$$

Let $C_1: r = 2\cos\theta$ and $C_2: r = \sqrt{3}$ be the polar curves

Find the area inside C_1 and outside C_2 .

Fall 2017

$$r = 2\cos\theta \quad r = \sqrt{3}$$

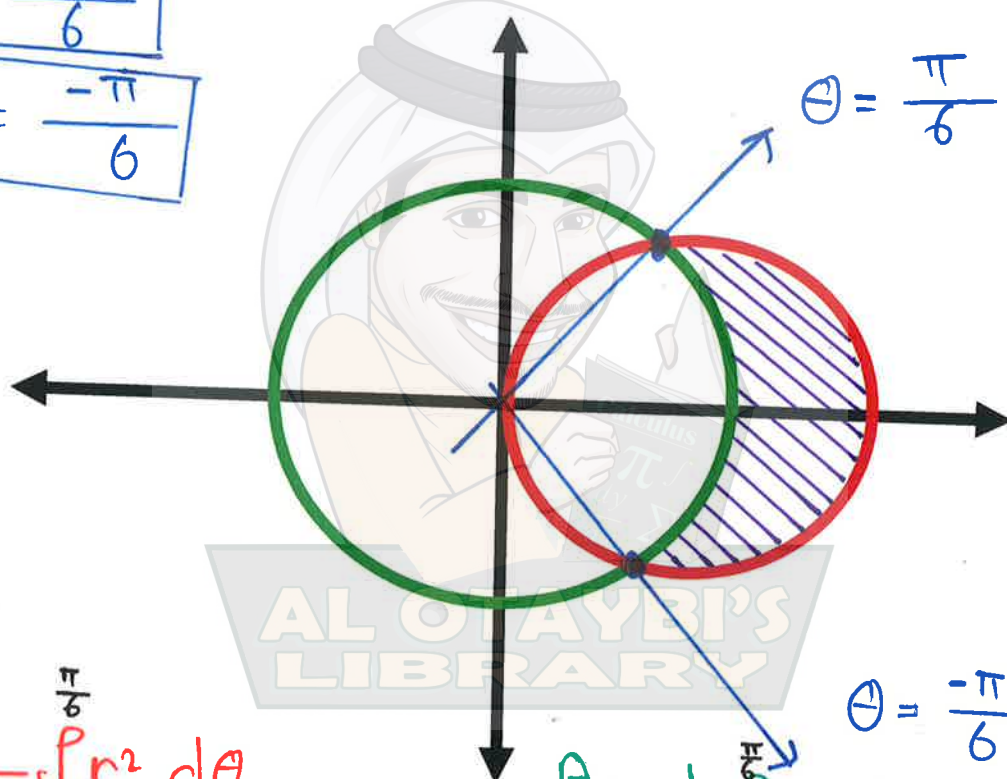
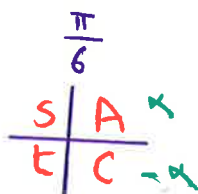
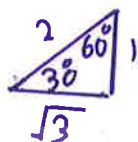
$$r = r$$

$$2\cos\theta = \sqrt{3}$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\theta = -\frac{\pi}{6}$$



$$\begin{aligned} A_1 &= \frac{1}{2} \int_{-\pi/6}^{\pi/6} r^2 d\theta \\ &= \frac{1}{2} \int_{-\pi/6}^{\pi/6} (2\cos\theta)^2 d\theta \\ &= \frac{1}{2} \cdot 4 \int_{-\pi/6}^{\pi/6} \cos^2\theta d\theta \end{aligned}$$

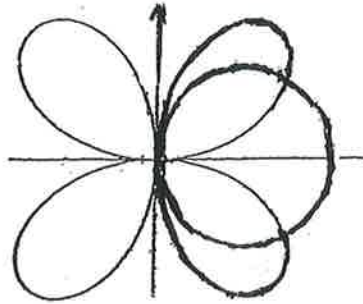
$$\begin{aligned} &= 2 \int_{-\pi/6}^{\pi/6} (1 + \cos 2\theta) d\theta \\ &= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/6}^{\pi/6} = 2 \left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} - 0 \right) \end{aligned}$$

$$\begin{aligned} A_2 &= \frac{1}{2} \int_{-\pi/6}^{\pi/6} r^2 d\theta \\ &= \frac{1}{2} \int_{-\pi/6}^{\pi/6} (\sqrt{3})^2 d\theta \\ &= \frac{3}{2} \left[\theta \right]_{-\pi/6}^{\pi/6} = \frac{3}{2} \left[\frac{\pi}{6} - 0 \right] \end{aligned}$$

$$A_T = 2(A_1 - A_2) = 2 \left(2 \left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} \right) - \frac{3}{2} \frac{\pi}{6} \right) = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

The figure below shows the graphs of the polar equations $r = \cos\theta$ and $r = \sin 2\theta$

Find the area of the region that lies inside both curves.



$$r = \cos\theta$$

$$r = \sin 2\theta$$

$$r = r$$

$$\cos\theta = \sin 2\theta$$

$$\cos\theta = 2\sin\theta \cos\theta$$

$$2\sin\theta \cos\theta - \cos\theta = 0$$

$$\cos\theta [2\sin\theta - 1] = 0$$

$$\cos\theta = 0$$

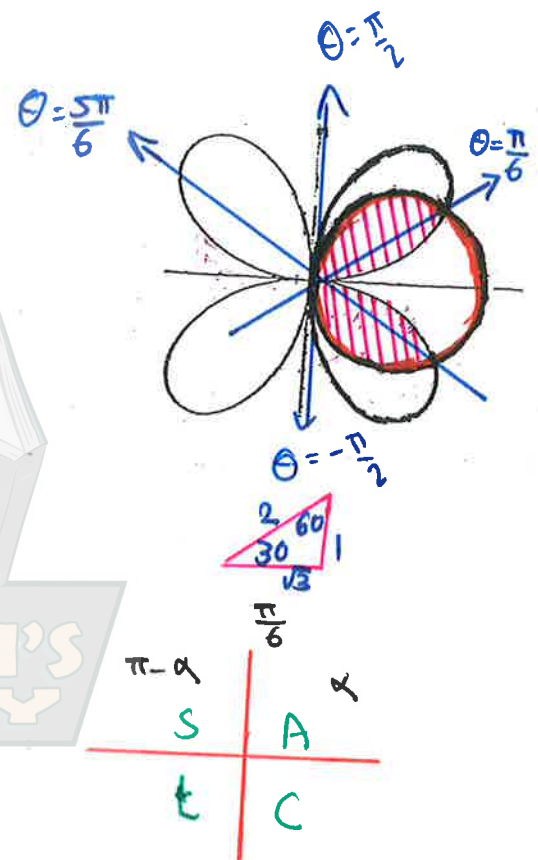
$$\sin\theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\Rightarrow \theta = \frac{-\pi}{2}$$

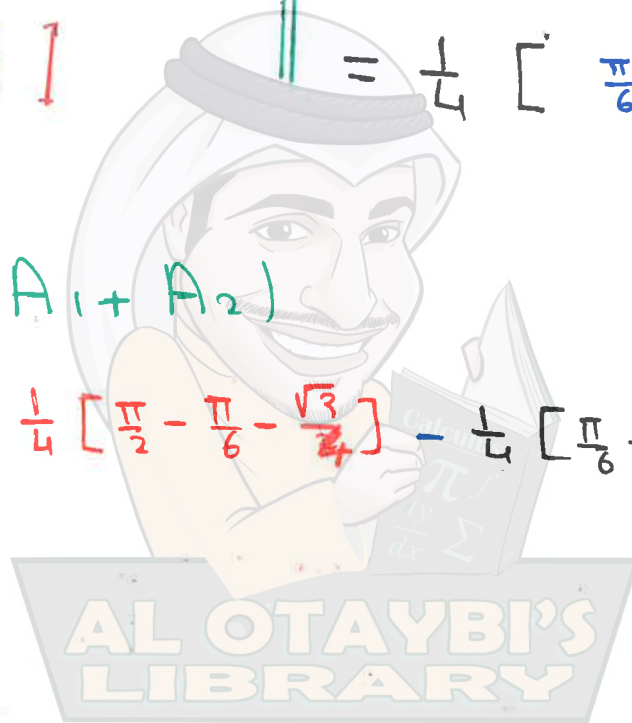
$$\Rightarrow \theta = \frac{5\pi}{6}$$



$$\begin{aligned}
 A_1 &= \frac{1}{2} \int_{\pi/6}^{\pi/2} r^2 d\theta \\
 &= \frac{1}{2} \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta \\
 &= \frac{1}{2} \cdot \frac{1}{2} \int_{\pi/6}^{\pi/2} (1 + \cos 2\theta) d\theta \\
 &= \frac{1}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{\pi/6}^{\pi/2} \\
 &= \frac{1}{4} \left[\left(\frac{\pi}{2} + 0 \right) - \left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} \right) \right] \\
 &= \frac{1}{4} \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{2} \right]
 \end{aligned}$$

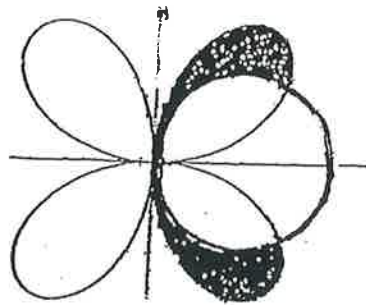
$$\begin{aligned}
 A_2 &= \frac{1}{2} \int_0^{\pi/6} r^2 d\theta \\
 &= \frac{1}{2} \int_0^{\pi/6} \sin^2 2\theta d\theta \\
 &= \frac{1}{2} \cdot \frac{1}{2} \int_0^{\pi/6} (1 - \cos 4\theta) d\theta \\
 &= \frac{1}{4} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{\pi/6} \\
 &= \frac{1}{4} \left[\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) - 0 \right] \\
 &= \frac{1}{4} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right]
 \end{aligned}$$

$$\begin{aligned}
 A_T &= 2(A_1 + A_2) \\
 &= 2 \left(\frac{1}{4} \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{2} \right] - \frac{1}{4} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right] \right)
 \end{aligned}$$



The figure below shows the graphs of the polar equations
 $r = \sin 2\theta$ and $r = \cos \theta$

Find the area of the shaded region



inside $r = \sin 2\theta$

outside $r = \cos \theta$

$$r = r$$

$$\sin 2\theta = \cos \theta$$

$$2\sin \theta \cos \theta = \cos \theta$$

$$2\sin \theta \cos \theta - \cos \theta = 0$$

$$\cos \theta [2\sin \theta - 1] = 0$$

$$\cos \theta = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

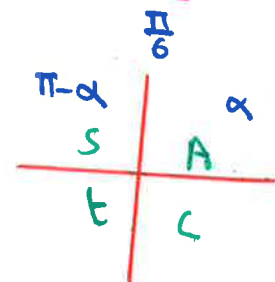
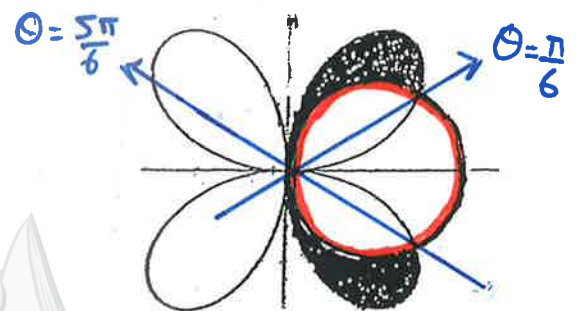
$$\Rightarrow \theta = -\frac{\pi}{2}$$

$$2\sin \theta - 1 = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\Rightarrow \theta = \frac{5\pi}{6}$$



$$A_1 = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin^2 2\theta d\theta$$

$$= \frac{1}{2} \cdot \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (1 - \cos 4\theta) d\theta$$

$$= \frac{1}{4} \left[\theta - \frac{\sin 4\theta}{4} \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \left[\left(\frac{\pi}{2} - 0 \right) - \left(\frac{\pi}{6} - \left(-\frac{\sqrt{3}}{4} \right) \right) \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right]$$

$$A_2 = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$= \frac{1}{2} \cdot \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{4} \left[\left(\frac{\pi}{2} + 0 \right) - \left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} \right) \right]$$

$$= \frac{1}{4} \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right]$$

$$\Rightarrow A_T = 2(A_1 - A_2)$$

$$= 2 \left[\frac{1}{4} \left(\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{8} \right) - \frac{1}{4} \left(\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) \right]$$

$$= 2 \cdot \frac{1}{4} \left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{8} - \frac{\pi}{2} + \frac{\pi}{6} + \frac{\sqrt{3}}{4} \right]$$

$$= \frac{1}{2} \left[\frac{-\sqrt{3}}{8} + \frac{2\sqrt{3}}{8} \right]$$

$$= \frac{1}{2} \left[\frac{\sqrt{3}}{8} \right] = \frac{\sqrt{3}}{4}$$

Setup the area of the region that lies inside the polar curve

$r = 1 - \sin\theta$ and outside the polar curve $r = \sin\theta$

$$r = 1 - \sin\theta$$

$$r = \sin\theta$$

$$r = r$$

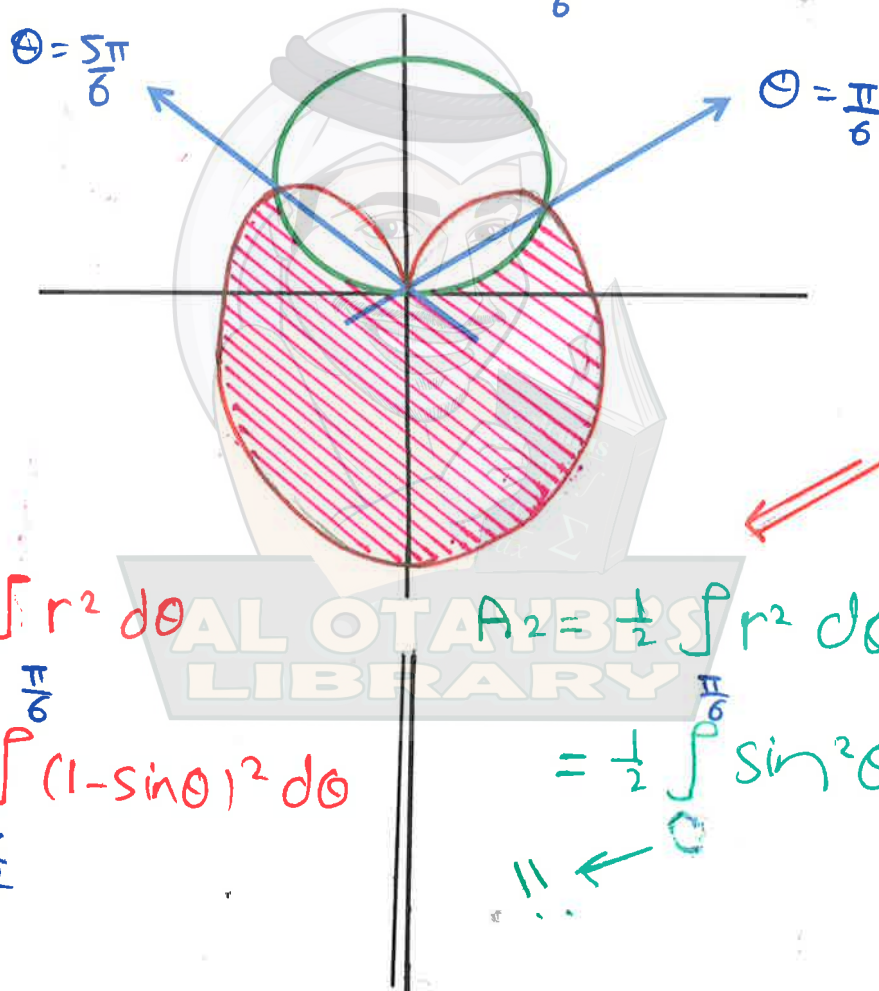
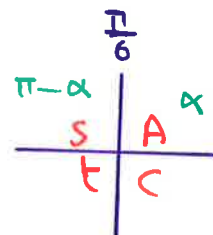
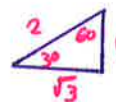
$$1 - \sin\theta = \sin\theta$$

$$\downarrow$$

$$1 = 2\sin\theta \longrightarrow \sin\theta = \frac{1}{2}$$

$$\implies \theta = \frac{\pi}{6}$$

$$\implies \theta = \frac{5\pi}{6}$$



* Trick *

$$A_1 = \frac{1}{2} \int r^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} (1 - \sin\theta)^2 d\theta$$

$$A_2 = \frac{1}{2} \int r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{6}} \sin^2\theta d\theta$$

!!

$$\implies A_T = 2(A_1 - A_2)$$

Find the area of the region that lies inside the circle $r = 2$
and above the horizontal line $r = \csc \theta$

Fall 2018

$$r = 2$$

$$r = \csc \theta$$

$$r = r$$

$$2 = \csc \theta$$

$$2 = \frac{1}{\sin \theta}$$

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

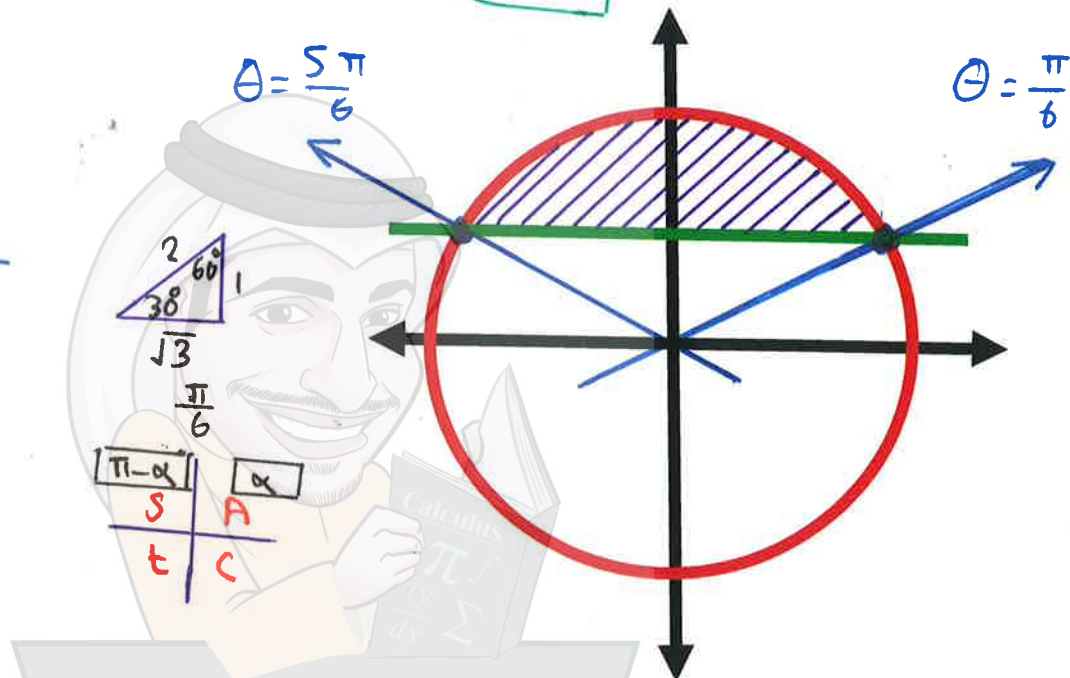
$$\theta = \pi - \frac{\pi}{6}$$

$$\theta = \frac{5\pi}{6}$$

$$r = \frac{1}{\sin \theta}$$

$$r \sin \theta = 1$$

$$y = 1$$



$$A_1 = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} r^2 d\theta$$

$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (2)^2 d\theta$$

$$= \frac{1}{2} 4 \left[\theta \right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$$

$$= 2 \left[\frac{5\pi}{6} - \frac{\pi}{6} \right]$$

$$= 2 \left[\frac{4\pi}{6} \right] = \frac{4\pi}{3}$$

$$A_2 = \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} r^2 d\theta$$

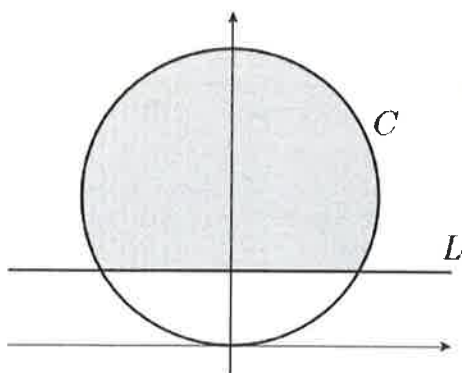
$$= \frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \csc^2 \theta d\theta$$

$$= \frac{-1}{2} \left[\cot \theta \right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$$

$$= \frac{-1}{2} \left[0 - \sqrt{3} \right] = \frac{\sqrt{3}}{2}$$

$$A_T = 2(A_1 - A_2) = 2\left(\frac{4\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

Let C be the circle $r = 4\sin\theta$ and L be the line $r = \csc\theta$ shown below



(a) Show that C and L are both symmetric about the vertical line $\theta = \pi/2$ (the y - axis).

(b) Find the area of the shaded region.

Jan 2017

$$r_1 = 4\sin\theta \quad r = \csc\theta$$

$$r_1(\pi - \theta) = 4\sin(\pi - \theta) = 4[\sin(\pi)\cos\theta - \cos\pi\sin\theta] = 4\sin\theta = r_1$$

$\therefore r_1$ is symmetric about $\theta = \frac{\pi}{2}$

$$r_2(\pi - \theta) = \csc(\pi - \theta) = \frac{1}{\sin(\pi - \theta)} = \frac{1}{\sin\pi\cos\theta - \cos\pi\sin\theta} = \frac{1}{\sin\theta} = \csc\theta = r_2$$

$\therefore r_2$ is symmetric about $\theta = \frac{\pi}{2}$

$$r = 4\sin\theta$$

$$r = \csc\theta$$

$$r = r$$

$$4\sin\theta = \csc\theta$$

$$4\sin\theta = \frac{1}{\sin\theta}$$

$$4\sin^2\theta = 1$$

$$\sin^2\theta = \frac{1}{4}$$

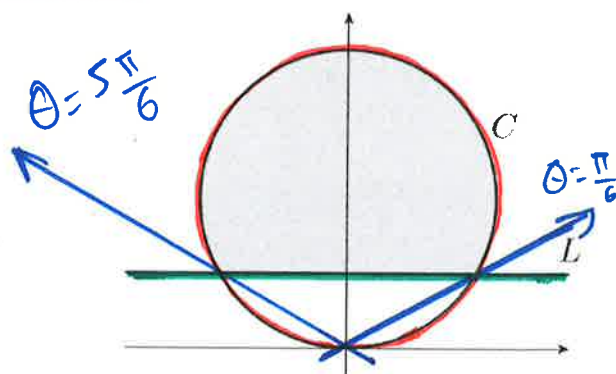
$$\sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

$$\sin\theta = \frac{-1}{2}$$

rejected



$$A_1 = \frac{1}{2} \int_{\pi/6}^{5\pi/6} r^2 d\theta$$

$$= \frac{1}{2} \int_{\pi/6}^{5\pi/6} (4\sin\theta)^2 d\theta$$

$$A_2 = \frac{1}{2} \int_{\pi/6}^{5\pi/6} r^2 d\theta$$

$$= \frac{1}{2} \int_{\pi/6}^{5\pi/6} \csc^2\theta d\theta$$

$$A_T = 2(A_1 - A_2)$$

Find the area of the region enclosed by the upper part of the cardioid

$$r = 1 - \sin\theta$$

Fall 2018

$$r = 1 - \sin\theta$$

$$A = 2 \cdot \frac{1}{2} \int_0^{\frac{\pi}{2}} r^2 d\theta$$

Symmetry

$$= \int_0^{\frac{\pi}{2}} (1 - \sin\theta)^2 d\theta$$

$$= \int_0^{\frac{\pi}{2}} (1 - 2\sin\theta + \sin^2\theta) d\theta$$

$$\frac{1}{2}(1 - \cos 2\theta)$$

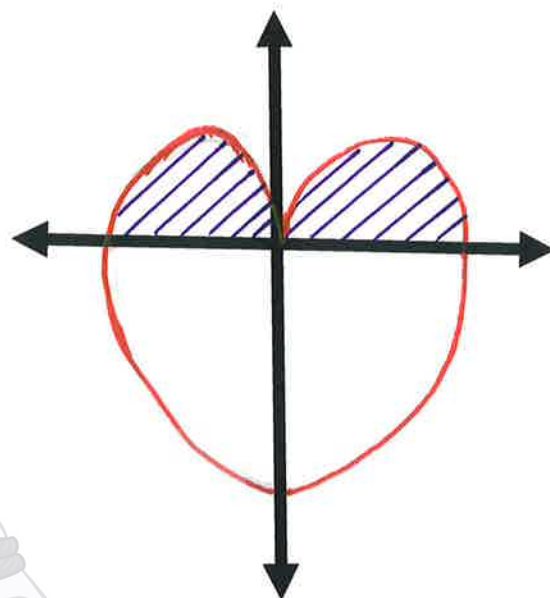
$$= \left[\theta + 2\cos\theta + \frac{1}{2} \left(\theta - \frac{\sin 2\theta}{2} \right) \right]_0^{\frac{\pi}{2}}$$

$$= \left[\frac{\pi}{2} + 0 + \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) \right] - \left[0 + 2 + \frac{1}{2} (0 - 0) \right]$$

$$= \frac{\pi}{2} + \frac{\pi}{4} - 2$$

$$= \frac{2\pi}{4} + \frac{\pi}{4} - 2$$

$$= \frac{3\pi}{4} - 2$$





Review



Let $c_1 : r = 3\cos\theta$, $c_2 : r = 1 + \cos\theta$

- Sketch the graph of both curves by using the same polar coordinate system.
- Set up the area of the region inside both c_1 and c_2 .
- Set up the area of the region inside c_1 and outside c_2 .
- Set up the area of the region inside c_2 and outside c_1 .

a) $r = 3\cos\theta$

$r = 1 + \cos\theta$

$r = r$

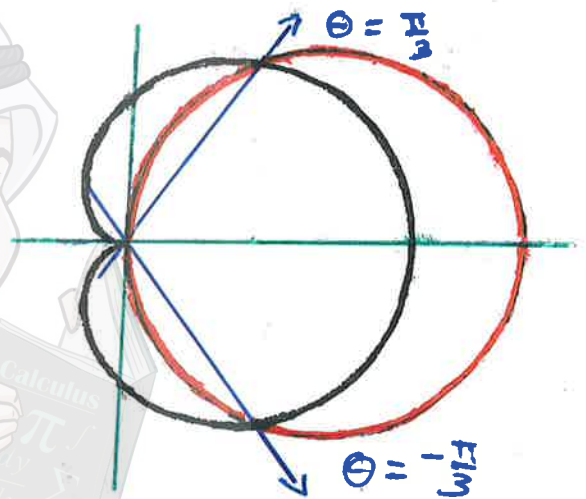
$3\cos\theta = 1 + \cos\theta$

$2\cos\theta = 1$

$\cos\theta = \frac{1}{2}$

$\Rightarrow \theta = \frac{\pi}{3}$

$\Rightarrow \theta = -\frac{\pi}{3}$



$A_1 = \frac{1}{2} \int r^2 d\theta$
 $= \frac{1}{2} \int (3\cos\theta)^2 d\theta$

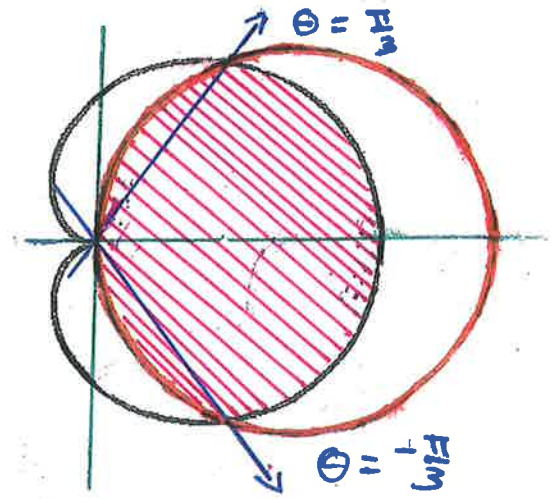
$A_2 = \frac{1}{2} \int r^2 d\theta$
 $= \frac{1}{2} \int (1 + \cos\theta)^2 d\theta$

$\Rightarrow$ See next page plz !

$$b) A_1 = \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (3 \cos \theta)^2 d\theta$$

$$A_2 = \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos \theta)^2 d\theta$$

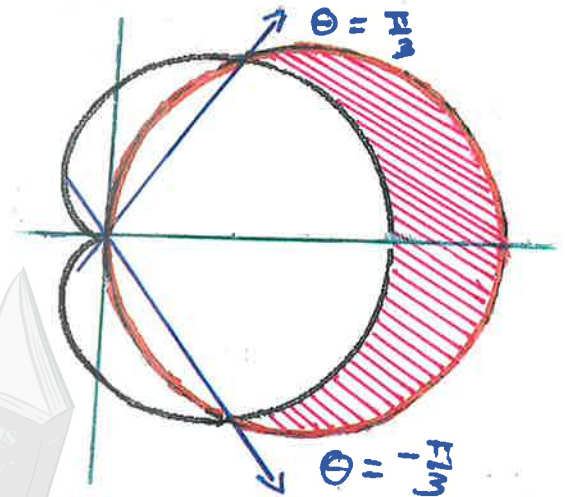
$$\Rightarrow A_T = 2(A_1 + A_2)$$



$$c) A_1 = \frac{1}{2} \int_0^{\frac{\pi}{2}} (3 \cos \theta)^2 d\theta$$

$$A_2 = \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos \theta)^2 d\theta$$

$$\Rightarrow A_T = 2(A_1 - A_2)$$

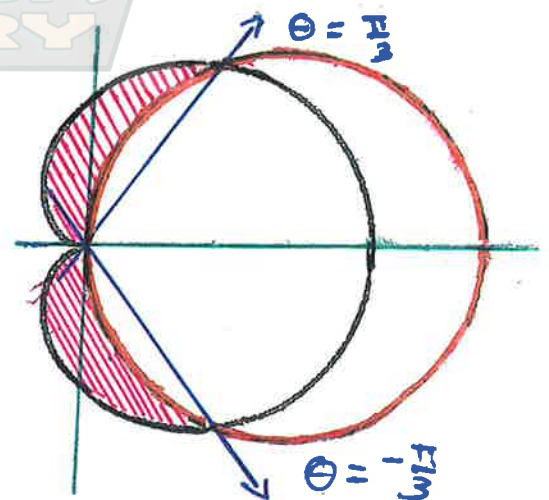


$$d) A_1 = \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} (3 \cos \theta)^2 d\theta$$

$$A_2 = \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} (1 + \cos \theta)^2 d\theta$$

$$\Rightarrow A_T = 2(A_2 - A_1)$$

inside



Consider the polar curve the polar curve

$$r = \theta - \frac{\pi}{4}, \quad \theta \in \left[0, \frac{\pi}{2}\right]$$

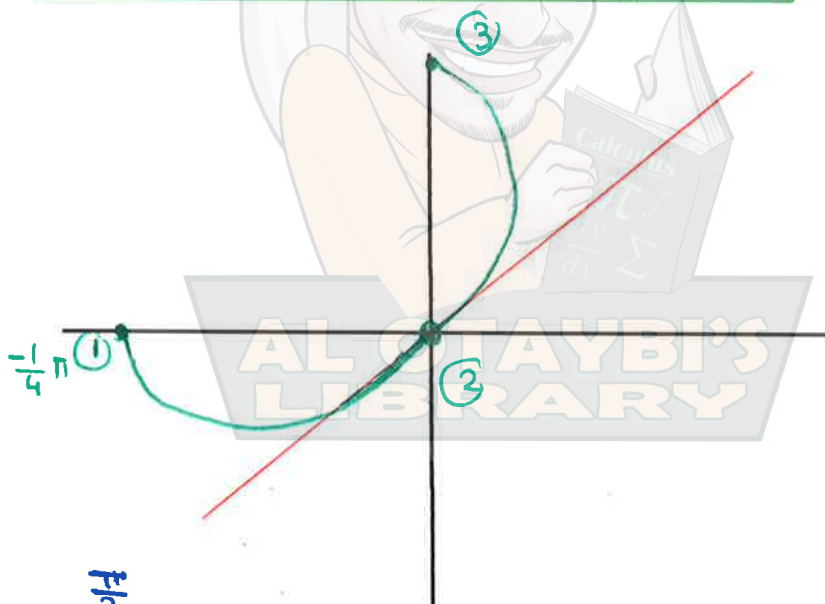
a) Plot the curve and label the points

corresponding to $\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$

b) Set up (but DO NOT evaluate the integral)
that represents the length of the curve.

$$r = \theta - \frac{\pi}{4}$$

θ	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
r	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$



$$L = \int_0^{\frac{\pi}{2}} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

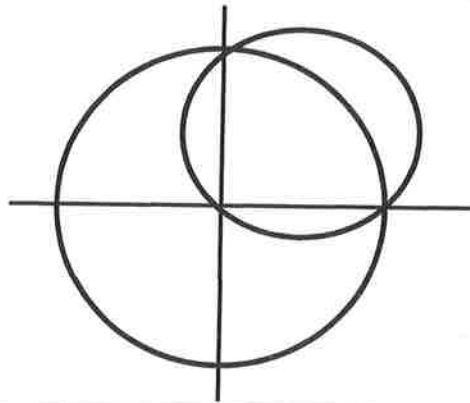
$$= \int_0^{\frac{\pi}{2}} \sqrt{\left(\theta - \frac{\pi}{4}\right)^2 + 1} d\theta$$

$$r = \theta - \frac{\pi}{4} \rightarrow r^2 = \left(\theta - \frac{\pi}{4}\right)^2$$

$$\frac{dr}{d\theta} = 1 \rightarrow \left(\frac{dr}{d\theta}\right)^2 = 1$$

$$\Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 = \left(\theta - \frac{\pi}{4}\right)^2 + 1$$

The graph of the polar equations $r = \sin\theta + \cos\theta$ and $r = 1$ as shown below. Find the area inside both circles



$$r = \sin\theta + \cos\theta$$

$$r = 1$$

$$r = r$$

$$\sin\theta + \cos\theta = 1$$

$$\theta = 0 \quad \theta = \frac{\pi}{2}$$

$$A_1 = \frac{1}{2} \int r^2 d\theta$$

$$= \frac{1}{2} \int (\sin\theta + \cos\theta)^2 d\theta$$

$$= \frac{1}{2} \int (\sin^2\theta + \cos^2\theta + 2\sin\theta \cos\theta)$$

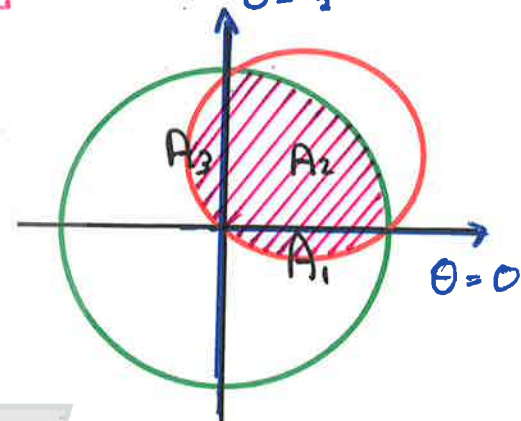
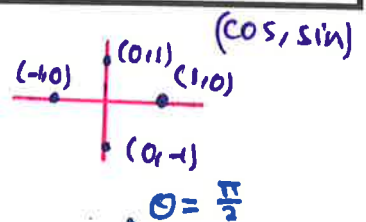
$$= \frac{1}{2} \int [1 + \sin 2\theta] d\theta$$

$$= \frac{1}{2} \left[\theta - \frac{\cos 2\theta}{2} \right]_{-\frac{\pi}{4}}^0$$

$$= \frac{1}{2} \left[(0 - \frac{1}{2}) - (-\frac{\pi}{4} - 0) \right]$$

$$= \frac{1}{2} \left[-\frac{1}{2} + \frac{\pi}{4} \right]$$

نستو sin الي اجهتها
لك cos ونطيقني
1

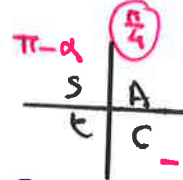


$$0 = \sin\theta + \cos\theta \quad | = \sin\theta + \cos\theta$$

$$\div \cos\theta$$

$$0 = \tan\theta + 1$$

$$\tan\theta = -1$$



$$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\theta = -\frac{\pi}{4}$$

$$\begin{aligned}
 A_2 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} r^2 d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1)^2 d\theta \\
 &= \frac{1}{2} [\theta]_0^{\frac{\pi}{2}} = \frac{1}{2} \left[\frac{\pi}{2} - 0 \right] = \frac{\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 A_3 &= \frac{1}{2} \int r^2 d\theta \\
 &= \frac{1}{2} \int (\sin \theta + \cos \theta)^2 d\theta \\
 &= \frac{1}{2} \left[\theta - \frac{\cos 2\theta}{2} \right]_{\frac{\pi}{2}}^{\frac{3\pi}{4}} \\
 &= \frac{1}{2} \left[\left(\frac{3\pi}{4} - 0 \right) - \left(\frac{\pi}{2} + \frac{1}{2} \right) \right] \\
 &= \frac{1}{2} \left[\frac{3\pi}{4} - \frac{2\pi}{4} - \frac{1}{2} \right] = \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow A_T &= A_1 + A_2 + A_3 \\
 &= -\frac{1}{4} + \frac{\pi}{8} + \frac{\pi}{4} + \frac{\pi}{8} - \frac{1}{4} \\
 &= -\frac{2}{4} + \frac{\pi + 2\pi + \pi}{8} \\
 &= -\frac{1}{2} + \frac{4\pi}{8} = -\frac{1}{2} + \frac{\pi}{2}
 \end{aligned}$$

Sol #2

$$\begin{aligned}
 A_T &= A_1 + A_2 + A_3 \\
 &= 2A_1 + A_2
 \end{aligned}$$

Sol #3

$$A_2 = \frac{1}{4} \pi r^2 = \frac{1}{4} \pi (1)^2 = \frac{1}{4} \pi$$

مساحة ربع دائرة